Correcting Control in “Common Ownership around the World”: Data and Exhibits

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Contents

1 Executive summary

This document accompanies two datasets built to correct the control side of the profit weights κ in "Common Ownership around the World." The first reconstructs control chains from Bureau van Dijk ownership filings across the 49 covered markets, tracing pyramids through intermediate holding layers to ultimate owners. The second corrects dual-class voting rights across 37 markets outside the United States by parsing each jurisdiction's own voting-rights disclosures with register-specific adapters that map statutes, prospectuses, annual reports, and regulatory filings into a common per-holder schema. The most developed instance derives US voting control from SEC DEF 14A proxy statements because US ownership disclosures (13F, 13D/G, and insider filings) report holdings rather than per-holder voting power by class and because the multiple-class firms themselves are excluded from the co-authors' panel. Voting control for those firms therefore had to be computed from the proxies' beneficial-ownership tables, charter terms, and footnotes. Together they supply corrected control vectors for 54,488 firms and change them for 20,842 (38.3%), roughly half of covered market capitalisation in the average market; at 2023Q4 the headline series applies a corrected vector to 27,174 of the co-authors' Thomson Reuters (TR) panel firms (72.2%), against 27,690 (73.6%) under the demoted permissive variant.

The exhibits display three series that differ only in which graded records they admit. *The headline series*[1] retains the discovered-controller records accepted by the grading, namely the *verified* and *unverified vendor-sourced* controllers, and holds out those it cannot corroborate;[2] *the strict series* admits only the verified tier; and *the demoted permissive variant*[3], the former shipped estimator, additionally retains the held-out records. Under the headline series the median market barely moves, while individual markets move in both directions; the correction that appears large is confined to the demoted permissive variant.

- Under the headline series the median market moves less than one percent: the all-market median change in $\bar{\kappa}$ is −0.83% at 2024Q4 (−0.76% from the control chains alone) and −0.29% at the composition-stable 2023 annual mean. The movement varies in sign across markets: down in controlling-shareholder markets (Korea −18.1%, Mexico −16.5%, Taiwan −15.4%, Malaysia −14.1%, Sweden −12.4%, Brazil −12.7%, New Zealand −12.1%), up in 14 markets, and negligible in Russia and Saudi Arabia.

- The United States correction changes sign as chain tracing reassigns votes toward identified controllers without removing links, so $\bar{\kappa}$ *rises* from 0.3525 (baseline) to 0.3606 at 2024Q4, a +2.3% change (44th of 49 markets by signed movement). The strict series, which admits only verified discovered controllers, moves the all-market median +0.71%, so the median sits just above zero under the most conservative reading.

- The large declines occur under the demoted permissive variant, which retains every discovered-

[1] Internally the MAIN∪SECONDARY control assignment, after the two record tiers it admits.

[2] *Held out* means excluded from every displayed series and kept at the co-authors' baseline vector. The held-out records are not all rejected: of their mass at 2024Q4, 38.0% is affirmatively refuted (custodians recorded as owners, stale entries, garbled names) and 62.0% is unresolved, representing potentially genuine controllers whose identity could not be established against the co-authors' roster, parked for later verification.

[3] Internally *combined_trace_DC_R1*; this is the estimator shipped in earlier drafts.

controller record including the held-out ones and moves the median −38.26%; the attribution exhibit assigns nearly all of that movement to the held-out records the headline series drops. The exhibits retain and explain that variant as a labeled series outside the headline.

- The headline is a known-direction lower bound with respect to newly discovered controllers because a record that fails verification but is not affirmatively refuted is held at the co-authors' baseline vector, leaving any correction it would carry outside the headline until later verification promotes it; the unverified but potentially genuine correction that remains parked is disclosed by country.
- The rise in common ownership survives the correction across the tabulated markets. The corrected United States series grows 401% against a 425% baseline while remaining above the baseline in level, the pyramid group's rise steepens (135% against 130%), and Japan's is preserved. Where the baseline rise was small the correction moves it modestly without reversing it: Korea's +1.9% becomes +4.7%, Australia's +9.4% becomes +3.0%, and the dispersed-group average of +5.4% becomes +4.0%.
- The DEF 14A derivation documents 360 genuine-wedge firms among the US multiple-class issuers the co-authors' panel excludes; among the 353 firms with a positive computed wedge, the median controller holds 33.9 percentage points more of the votes than of the cash flows, with a maximum separation of 93.9 points.
- The multiclass exclusion itself biases the shipped panel: the excluded firms are exactly those whose share structures separate control from ownership, and restoring them shows the exclusion overstates the panel-average level of $\bar{\kappa}$ while hiding real common-ownership linkage among the firms that remain. For the United States at 2024Q4 the average $\bar{\kappa}$ moves from 0.3525 (baseline) to 0.3606 under the headline series on the shipped universe and to 0.2301 once the 946 reinstated firms enter; the reinstatement dilutes the incumbent average (−0.0846) while adding genuine common-ownership linkage (+0.0168, positive in all twenty quarters). The correction is therefore presented on both universes, so the exclusion bias is read directly as the difference between them. The exclusion is panel-wide (3,620 issuers across 41 markets, censused in Section 8), where the complete disposition of every excluded firm is now reported. The certified append spans twenty Panel-B markets (138 firms) and the 946 United States nodes, two tagged non-citable variants add a further 216 firms, and the rollout to the remaining markets continues.
- Each correction channel is also reported in isolation, so either data layer can be quoted without the other: the control-chain channel carries nearly the entire in-panel movement (active in 47 of 49 markets), while the dual-class channel is an order of magnitude smaller and operates mainly through the reinstated firms.

The headline series rests on record-level grading. Beyond re-weighting holdings that TR already records, the chain tracing sometimes discovers a controller that TR does not carry. Each discovery is a single candidate record graded by the strength of its evidence into the verified, unverified vendor-sourced, and held-out tiers defined in Section 2. Rebuilding $\bar{\kappa}$ across these grades gives the range the attribution exhibit reports, from +0.71% at the median with verified records only and −0.76% adding the unverified vendor tier (control chains alone) to −38.26% for the demoted permissive variant. The estimator choice is therefore recorded as taken, with the headline series set as the verified-plus-unverified series pending co-author ratification. Sections 3 and 10 document the grading discipline and attribution evidence.

Robustness checks show little sensitivity to the modelling rules. The pyramid-propagation rule is immaterial for the full corrected series (median absolute difference 0.0001 in $\bar{\kappa}$), and the dual-class attribution rule is immaterial outside two markets (Brazil and Sweden) in the standalone dual-class satellites; the loyalty quarantine moves the former Italian and Dutch sensitivity into the loyalty satellite comparison.

1.1 Roadmap

Section 2 introduces the control correction; the exhibits then follow as Sections 3 through 12, ordered so that the data are described before their consequences for κ are drawn. Sections 3 through 5 construct the two correction datasets, summarise them, and give the country-by-country coverage census; Sections 6 through 9 report corrected levels, the time-series results, the reinstatement of the excluded multiclass firms, and the excluded US universe from the DEF 14A proxies; and Sections 10 through 12 decompose the correction by record grade, vary the two consequential modelling rules, and list the open decisions and pending work. Each section is self-contained, with its own tables, figures, and source notes.

2 Overview of the control correction

2.1 The proportional-control assumption

The co-authors' profit weights κ are computed from Thomson Reuters (TR) institutional holdings under a proportional-control assumption: a holder's control over a firm equals its cash-flow stake, one share carrying one vote. Although that description is adequate for a widely held firm without a dominant blockholder, it does not describe firms in which control exceeds cash-flow rights. In a pyramid, a family or a state entity controls a listed firm through intermediate holding companies, and the controller's voting power at each layer can substantially exceed the cash-flow claim obtained by multiplying stakes down the chain. Under a dual-class charter, a super-voting share class can similarly concentrate votes in a small economic position. In either case, proportional-control κ assigns too much influence to institutional investors and omits the common-ownership links created by the controller's other holdings. The exhibits document a correction to the control side of κ for these firms and describe its data and empirical implications.

2.2 Control-chain and dual-class data

The correction uses two new datasets, whose full construction census is reported in Section 4. The first reconstructs control chains from Bureau van Dijk ownership filings, tracing 755,944 firm–owner links across the 49 covered markets to ultimate owners and identifying a controlling owner (most often a named family or individual) for 48,344 of the 72,807 listed firms. The second corrects dual-class voting rights for 1,697 candidate multiple-class firms in 37 markets outside the United States, of which 893 carry a genuine voting-power wedge, and for the US multiple-class issuers the co-authors' panel excludes altogether. Their voting control is derived from SEC DEF 14A proxy statements, and 360 of the 702 in-scope candidates resolve to a wedge.

2.3 Discovered-controller records and grading

The correction first re-weights holdings that TR already records when the traced control structure reassigns the firm's votes; this component rests entirely on holdings the paper already trusts. The ownership chains also sometimes reveal a controller that TR does not carry, such as a family, founder, or state entity holding a large stake in some company. Each such discovery is a single record asserting that one particular owner controls one particular company; across the 49 markets, the correction generates 47,392 candidate records of this kind.

The grading of every candidate record determines which records the headline series carries. The *verified discovered controllers* (165 records) were each confirmed against primary sources and have survived repeated adversarial audits. The *unverified vendor-sourced controllers* (1,381 records) come from the ownership vendor's chain data, and a fresh audit sample measured their error rate at 26.0% (one-sided 95% upper bound 38.1%). The remaining 45,846 candidates could not be corroborated and are held out of the headline; of their mass at 2024Q4, 38.0% is affirmatively refuted (custodians recorded as owners, stale entries, garbled names) and 62.0% is unresolved and potentially genuine, parked for later verification. These grades attach to individual records, never to firms or countries; a single firm can carry a verified record alongside a held-out one. The headline series retains the verified and unverified records and adds a synthetic cash-flow (denominator) leg for each; the demoted permissive variant additionally retains the 45,846 held-out records, and the strict series admits only the verified tier. The choice of which series the paper ultimately ships is recorded as taken pending co-author ratification. The exhibits present the headline series alongside the strict series and the demoted permissive variant, with the attribution evidence for the choice.

2.4 The substitution overlay

The correction enters the co-authors' computation as a substitution overlay. For each treated firm the control vector γ is replaced by the corrected one, with the holdings data and the other ingredients of the panel retained; untreated firms remain unchanged. On the cash-flow side, the TR vector β is kept as recorded for holdings TR already carries; the headline series additionally writes a synthetic cash-flow (denominator) leg for each discovered controller absent from TR, so only the strict series retains the TR cash-flow data verbatim. The overlay computes corrected control vectors for 54,488 firms and changes them for 20,842 (38.3%); in the average market the treated firms account for about half of covered market capitalisation.

Measured common ownership depends on the control assumption used to construct it; the estimator choice is recorded as taken pending co-author ratification, and the exhibits that follow report the results on levels, dynamics, provenance, and robustness.

2.5 Series, bases, and caveats

The exhibits share the following series, reporting bases, and caveats, stated here once and referenced by the individual exhibit notes. Three series are displayed throughout: the *headline series* (internally MAIN∪SECONDARY), admitting the verified and unverified vendor-sourced discovered controllers; the *strict series*, admitting the verified tier only; and the *demoted permissive variant* (*combined_trace_DC_R1*, the former shipped estimator), which additionally retains the held-out records. The record counts progress through three stages: the unverified vendor-sourced tier is graded at 1,381 records, of which 1,373 survive into the frozen decision cache (eight demoted to held out on the final rebuild) and 1,352 fall within the 2024Q4 panel roster, while the held-out tier correspondingly runs 45,846 graded, 45,854 cached, and 45,838 applied at 2024Q4; the verified tier holds 165 records. Of the held-out mass at 2024Q4, 38.0% is affirmatively refuted and 62.0% is unresolved and potentially genuine. Record grading is orthogonal to universe membership: the three series are computed on the co-authors' shipped firm universe, which excludes the multi-class firms of Section 8 even though their reinstated voting vectors are per-firm adjudicated, so a firm can be verified and nonetheless absent from every displayed series for a sample reason; exhibits that restore those firms are labelled enlarged and reported in Section 8. Levels are reported on two bases, the point-in-time 2024Q4 quarter and the composition-stable 2023 annual mean; because firms that have left the panel by 2024Q4 drop out of the applied counts, that quarter carries a roster-attrition caveat and the 2023 annual mean is the more stable basis for comparisons across time. All series are anchored to a single reconstruction vintage, whose per-market coverage and overfull profile are documented in Section 3. All numbers reported in these exhibits are non-citable pending the R3 cross-model audit.

3 Data construction

The two correction datasets (a control-chain reconstruction and a non-US dual-class panel), together with a separate US DEF 14A derivation, were built from primary and vendor sources across the 49 covered markets and audited before any number entered a κ computation. This section gives the design principle common to every pipeline (Section 3.1), a one-paragraph summary of each dataset and its verification (Sections 3.2–3.5), and the integration with the co-authors' panel (Section 3.6); the construction mechanics are deferred to Appendix B.

3.1 Deterministic processing and model-assisted judgment

The construction must be reproducible, with every number regenerable by committed code from archived inputs, while accommodating judgments that vary by jurisdiction and cannot be reduced to a universal rule. Examples include whether two names in different scripts denote one family and whether a register entry is a named holder or a dispersed pool. Hand-curated keyword lists cannot make such judgments because a list calibrated on the jurisdictions already inspected fails on the next. The pipelines therefore assign everything jurisdiction-invariant to a deterministic engine (the Shapley–Shubik power computation and voting cutoff, ownership-tree traversal, the pyramid cash-flow product, code parsing, and every screen and bounds check), while confining the judgment tier to jurisdiction-varying decisions grounded in a written per-jurisdiction brief.

The model never runs at build time: each decision is made offline and frozen into a provenance-tagged cache the engine consumes, so a rebuild reproduces every value and the caches become the population the audits sample. Four constraints bound the tier: closed-set outputs the engine can validate; conservative defaults that admit a controller only on positive evidence, so failure costs coverage rather than inventing numbers; consequence-ranking of decisions; and provenance tags making any decision reversible in isolation. Table 1 shows how this division applies to each task.

Table 1: Division of labour in the construction pipelines

Stage	Deterministic engine (runs at build time)	Judgment tier (decided offline, cached with provenance)
Acquisition	Vendor pulls and register harvests; immutable raw archives with checksums and pull dates	—
Extraction	Table walking, footnote-to-row linkage, locale and script handling, percentage and type-code parsing	Bounded extraction of narrow fragments where deterministic rules fail or report low confidence
Identity	Surname aggregation, ultimate-owner roll-up, vehicle merging by rule	Name-to-family keying; dispersed-pool classification; same-entity and concert adjudication
Control arithmetic	Shapley–Shubik and cutoff classification, pyramid products, denominators, wedges	—
Sample gating	Screens, bounds guards, quarantine of violating rows	Wedge-status adjudication of candidate multiple-class firms
Verification	Embedded assertions, named anchors, population-level movement guards	Adversarial audits; cross-model for the highest-stakes datasets

3.2 Control-chain reconstruction

The control-chain reconstruction begins from Bureau van Dijk (BvD) ownership filings, pulling each market's listed-firm universe and first-level shareholders and expanding every significant corporate holder recursively through the global ownership database, so a chain passes through a private holding company to the family or state entity behind it. On the consolidated blocks the engine computes control by the 20% voting cutoff and Shapley–Shubik pivotality (reconstructed from the worked examples in Aminadav and Papaioannou's appendix) and cash-flow rights β as the pyramid stake product capped at the control stake γ, so $\gamma - \beta$ measures pyramid attenuation. Owner identity is central, since one owner fragmented across nominees and holding vehicles would omit the cross-firm links κ depends on; the resulting control and family shares are validated country by country against those Aminadav and Papaioannou publish. The consolidation rules, the concert tier, the three identity passes, and the movement guards are given in Appendix B.

3.3 Dual-class voting rights outside the United States

The non-US dual-class dataset measures, from primary ownership registers scattered across dozens of national regimes with no common format, who holds which share class of each multiple-class firm and how many votes each class carries. Register-specific adapters map each regime into a common schema, from which the build computes voting power, cash-flow rights, and the wedge under fixed per-jurisdiction conventions: statutory votes-per-share from law where it fixes them, par-weighted cash-flow legs where votes attach to nominal capital, and loyalty-share regimes flagged as a satellite set. Because a class-count screen abroad is unreliable, every candidate was adjudicated before entering the panel: of 1,697 candidate multiple-class firms across 37 markets, 893 carry a genuine voting-power wedge, about half of those a screen flags. Because nothing is imputed, an unpublished leg stays empty and a missing controller is never invented; the construction stages, the bounds and statute-envelope guards, the named controller anchors, and the round-trip gate against the US wedge set are given in Appendix B.

3.4 The United States: deriving voting control from DEF 14A proxy statements

The United States required a separate derivation, because the co-authors' panel excludes US multiple-class issuers outright and US proxy statements rarely report a holder's voting power directly. The pipeline derives it from the Item 403 beneficial-ownership share counts, the charter's votes-per-share terms, and the footnote attribution across affiliated trusts and vehicles, computing the wedge at the consolidated control-group level; a parser reading only the ownership table would miss the super-voting class through which a founder-and-trusts group's combined vote exceeds its economic stake. The universe is the excluded-issuer set bridged to EDGAR by CIK and collapsed to 946 representative CIK nodes, from which the derivation resolves 360 genuine wedges, every scoped firm leaving the funnel with a documented verdict. The three-tier extraction, the two denominator conventions, and the held-out golden benchmark are given in Appendix B.

3.5 Verification and reproducibility

Every dataset above was audited before its numbers were admitted downstream. The audit followed four rules: review independent of the build and conducted adversarially, with a model family different from the one used in construction checking the highest-stakes datasets against primary documents (the US derivation was audited cross-model against the proxy statements before it entered any κ computation); class-based remediation, in which each audit finding is enumerated over the full population and corrected as a class instead of case by case; record grading, which sorts the discovered-controller records into the evidence tiers defined in Section 2.5; and reproducibility, under which every exhibit number is generated by committed code from provenance-tagged artifacts. The overlap audit's de-duplication round moved 8 vendor-tier records into the held-out set, so the post-R15 cache the headline series consumes retains 1,373 vendor-sourced controllers (from the 1,381 graded) against 45,854 held out. The word *certified* remains reserved for individually verified records and named prior-campaign artifacts, never for a current series, basis, or result.

3.6 Integrating the correction with the co-authors' panel

The co-authors' panel already embeds its own hand corrections, including fund-family aggregation, BlackRock–BGI consolidation, a single consolidated Chinese state owner, and TR-internal family blocks. Because a discovered controller added naively could duplicate an adjustment already made in the panel, four guards prevent this. *Replacement*: a discovered controller matching a position TR carries replaces the matched conduit row. *Absence*: a controller with no TR counterpart is admitted only after a per-name check confirms its absence from the full holder population. *Conservative default*: a record that neither matches nor clears the absence check is quarantined, so ambiguity resolves toward the co-authors' baseline. The panel's showcased Hermès row (66.7%) stands alone, while a competing depth-two conduit fragment was quarantined and nothing added. *Arithmetic backstop*: any issuer whose retained synthetic legs plus TR-recorded cash-flow rights exceed unity reverts to baseline, swept across all 49 markets. In $\gamma \times \beta$ units at 2024Q4 the integration therefore mostly re-attributes rather than adds: 50.5% of discovered-controller mass substitutes for a TR position already present, 48.6% is quarantined and never enters $\bar{\kappa}$, and only 0.9% enters as genuinely new mass under the absence guard. China is the closest overlap case (their consolidated state owner against our tracing through provincial vehicles); Appendix B carries the vehicle-level detail.

Quarantine is a deliberate under-correction: an unverified record absent from TR that violates no guard stays out on its baseline vector until affirmative evidence promotes it, so the headline understates the controllers TR missed. Of the quarantined mass at 2024Q4, 38.0% is affirmatively refuted (custodians booked as owners, stale entries, garbled names), and 62.0% is unresolved, comprising potentially genuine controllers parked, reason-tagged, for later rounds. The cross-model overlap audit confirmed the design: the 65,915 matched assignments substitute rather than sum (1,213.77 → 1,234.14 in denominator mass), a full scan of the 1,381 unverified vendor-sourced records against current TR holders returned zero name collisions, the one confirmed hand-correction overlap (a Spanish family aggregate absorbing a spouse already in TR) was quarantined with seven other flagged records, and the demoted permissive series was confirmed to double-count the verified anchors against their retained TR rows. The backstop's firing profile (6 issuers at 2024Q4, 127 a quarter earlier, roughly 452 per quarter through 2019, with 25,896 issuer-quarters in all) measures the growing arithmetic incompatibility of the 2020–2024 control vintage with earlier cash-flow records, and is why the corrected headline is anchored in the 2020–2024 window.

4 Summary statistics for the correction data

The exhibits in this section describe the correction datasets before any profit weight is computed. Table 2 reports their contents, Table 3 the divergence between corrected control vectors and the cash-flow vectors they replace, and Table 4 the types of identified controllers. Table 20 in Appendix A reports their market-level overlap with the co-authors' Thomson Reuters (TR) holdings panel by market. Every number is computed from the audited pipeline artifacts; counts that also appear in the coverage census (exhibit T1) are machine-checked against it.

4.1 Construction census

The correction uses two newly constructed datasets. The first reconstructs control chains from Bureau van Dijk ownership filings and, across 49 markets, assigns a control status to 72,807 listed firms, tracing 755,944 firm–owner links through intermediate holding layers to ultimate owners. Two thirds of these firms (48,344, 66.4%) have an identified controlling owner, most often a named family or individual. The reconstruction yields 619,853 owner-level records with both a control leg γ and a cash-flow leg β for 54,248 firms; 72.7% of the records are extracted deterministically from the filings, with the remainder resolved through the vendor's ultimate-owner roll-up, nominee look-through, or a cached and provenance-tagged LLM adjudication of the hardest name-resolution cases.

The second dataset covers dual-class voting rights. Outside the United States, 1,697 candidate multiple-class firms in 37 markets were adjudicated against statutes, prospectuses, and annual reports; 893 (52.6%) carry a genuine voting-power wedge, and 616 firms in 22 markets enter the overlay with per-holder voting and cash-flow stakes. For the United States, where multiple-class firms are excluded from the co-authors' panel, voting control was derived directly from DEF 14A proxy statements. Of the 946 representative CIK nodes bridged from the excluded issuers, 702 are in-scope multiple-class candidates; the derivation resolves 360 genuine wedges, 261 equal-vote multiclass structures, and 66 single-class artifacts, leaving 15 documented as unresolvable.

4.2 Overlap with the co-authors' TR universe

Table 20 in Appendix A reports the overlap with the co-authors' TR holdings panel for each market. At the composition-stable reference quarter 2023Q4 the TR panel carries 37,639 firms across the 49 markets, while the correction datasets cover 54,488 firms; 36,793 covered firms (67.5%) match a TR issuer through the audited firm crosswalk (ISIN or name match). Because coverage is broader than the TR roster in most markets, the overlap is high from the panel's side. The demoted permissive variant reaches 27,690 TR-panel firms (73.6%) in the reference quarter, and the headline series reaches 27,174 (72.2%); the two differ by the 516 firm-quarters that record grading reverts to baseline. In the average market the applied firms hold 71.6% of TR-panel market capitalisation, so the correction reaches the bulk of the co-authors' universe by capitalisation even though the treated-firm count (20,842 firms whose control vector actually changes) is well below the applied count, since for many firms the corrected vector coincides with the baseline.

4.3 Control–cash-flow divergence before κ

Table 3 summarises the divergence between the corrected control vector γ and the cash-flow vector β among the 20,842 treated firms, by treatment channel. At the median treated firm, the largest owner-level $|\gamma - \beta|$ is 0.083, corresponding to 8.3 percentage points of control reassigned at a single owner. It rises to 0.375 at the 90th percentile, and the mean is 0.144. Divergence is larger in the dual-class channels than in the chains-only channel (median 0.172 for dual-class-only and 0.221 for both-channel firms, against 0.081); firms treated by both mechanisms show the largest

Table 2: Construction census of the correction datasets

	N	Markets	Share (%)
Panel A: Control-chain reconstruction (BvD ownership tracing)			
Listed firms with a reconstructed control assignment	72807	49	–
controlled firm: named family or individual	29480	–	40.5
controlled firm: private holding vehicle	15849	–	21.8
controlled firm: state	3015	–	4.1
widely held (with or without a blockholder)	24463	–	33.6
controlled through SS pivotality alone	5635	–	7.7
Ultimate-owner links traced (firm × owner)	755944	49	–
Firms entering the κ overlay universe	54248	49	–
Owner-level corrected records (γ, β legs)	619853	49	–
extracted deterministically	450639	–	72.7
vendor GUO roll-up	168335	–	27.2
nominee look-through	740	–	0.1
LLM-adjudicated (cached, provenance-tagged)	139	–	0.0
Panel B: Dual-class voting rights (non-US markets)			
Candidate multiple-class firms adjudicated	1697	37	–
genuine voting-power wedge	893	31	52.6
equal-vote multiclass	154	–	9.1
non-equity instrument artifact	226	–	13.3
board-representation wedge only	9	–	0.5
no usable disclosure	415	–	24.5
Firms entering the overlay with per-holder γ/β	616	22	–
Holder-level records	3854	22	–
Panel C: United States DEF 14A derivation (proxy statements)			
Excluded-universe CIK bridge	946	1	–
screened out (not multiple-class)	244	–	25.8
scoped multiple-class candidates	702	–	74.2
genuine voting–cash-flow wedge	360	–	51.3
equal-vote multiclass	261	–	37.2
single-class artifact	66	–	9.4
documented unresolvable	15	–	2.1

Note: Series, bases, and caveats as defined in Section 2.5. The table gives the construction census of the two correction datasets. Panel A reports the control-chain reconstruction from Bureau van Dijk ownership filings over the 49 covered markets (the reference-only Amadeus duplicate of France is excluded). A firm is *controlled* when its largest ultimate-owner block holds at least 20% of the voting rights or is Shapley–Shubik pivotal: 42,709 controlled firms meet the 20% criterion and 5,635, displayed separately, are controlled through pivotality alone; controller categories follow the Aminadav–Papaioannou typology. *Share* is a percentage of the immediately enclosing total, using firms for the typology rows and owner-level records for the extraction rows. Panel B reports the non-US dual-class adjudication (*wedge_status/verdicts.csv*); its share column gives percentages of adjudicated firms. Panel C reports the US DEF 14A derivation of voting control for issuers excluded from the co-authors' panel, with shares as percentages of the 946-node CIK bridge for the screening rows and of the 702 scoped candidates for the verdict rows.

mean and median divergence, while dual-class-only firms carry the heavier upper tail (75th and 90th percentiles). The signed $\gamma - \beta$ at the largest-divergence owner is positive for 99.6% of treated firms, indicating that the divergence is almost uniformly a gain of control over cash-flow rights. For the chains mechanism, this holds by construction because the pyramid cash-flow product is capped at the control stake, while the dual-class mechanism also admits inferior-voting blocks (76.6% positive). The correction also changes the identity of the largest controller at 3,673 treated firms (17.6%). Summed over all treated firms, the reallocated control mass amounts to 1,652 firm-control equivalents, of which 1,542 arise in the chains-only channel.

Table 3: Pre-κ divergence between corrected control and cash-flow rights, by channel

	Chains only	Dual class only	Both	All treated
Treated firms	20292	338	212	20842
Largest owner-level $\lvert\gamma - \beta\rvert$ per firm				
mean	0.142	0.218	0.256	0.144
25th percentile	0.031	0.048	0.131	0.031
median	0.081	0.172	0.221	0.083
75th percentile	0.199	0.351	0.347	0.203
90th percentile	0.370	0.506	0.492	0.375
Signed $\gamma - \beta$ at that owner				
mean	0.142	0.176	0.241	0.143
share positive (%)	100.0	76.6	94.3	99.6
Identity of the largest controller				
firms whose largest controller changes	3576	49	48	3673
share of treated firms (%)	17.6	14.5	22.6	17.6
Control mass reallocated				
mean $\sum_o \lvert\gamma_o - \beta_o\rvert$ per firm	0.152	0.345	0.494	0.159
aggregate, firm-control equivalents	1542	58	52	1652

Note: Series, bases, and caveats as defined in Section 2.5. The table reports the distribution of control–cash-flow divergence among the 20,842 treated firms, split by treating mechanism (chains only, dual class only, or both; disjoint and summing to the total, with definitions as in exhibit T1). The firm-level divergence is the largest owner-level $\lvert\gamma-\beta\rvert$ across the firm's owner rows, γ the traced control (chains) or voting-power (dual-class) stake and β the corresponding cash-flow stake, both firm shares in $[0, 1]$. *Signed* $\gamma - \beta$ takes the mechanism with the larger divergence, nonnegative by construction for chains (the pyramid cash-flow product is capped at the control stake) and either sign for dual class. The *largest controller changes* when the largest-γ and largest-β owners differ within the treating mechanism (the 20 dual-class rows with a missing cash-flow leg are excluded here only). *Control mass reallocated* is $\sum_o \lvert\gamma_o - \beta_o\rvert$ over the firm's overlay owner rows, halved in the aggregate row so one unit equals one firm's entire control changing hands; quantiles use the nearest-rank convention.

4.4 Controller types and record grades

Table 4 describes the controllers the reconstruction identifies. Named families and individuals are the modal controller worldwide, controlling 29,480 firms (40.5% of the 49-market universe), with a pronounced regional gradient, as they control 46.5% of firms in Asia–Pacific but 26.9% in the Americas, where widely-held firms without any blockholder are far more common (28.1% against 6.1%). Private holding vehicles, including both traced and opaque entities, control a further 15,849 firms (21.8%), and state control (3,015 firms, 4.1%) concentrates in Asia–Pacific and the Middle East.

Panel B grades the owner-level records that the reconstruction adds to the TR data. Each record asserts that a controller absent from the TR data holds at least 3% of a listed firm. Of the 47,392 audited candidates, 165 are verified discovered controllers confirmed against primary sources, 1,373 are unverified vendor-sourced controllers carrying a disclosed measured error rate, and 45,854 (96.8%) are held out of the headline (grade definitions and the refuted/unresolved split as in Section 2.5). The held-out mass is excluded from the headline and strict series but retained in full by the demoted permissive variant. The attribution exhibit measures the consequence of that inclusion. The note to Table 4 gives the 1,381 → 1,373 overlap-audit reconciliation and the per-record decomposition of the remaining overlay mass.

Table 4: Controller anatomy: typology of identified controllers and record tiers

	N	All (%)	Am. (%)	Eur. (%)	AP (%)	MEA (%)
Panel A: Firm-level controller typology, 49 markets						
Named family or individual	29480	40.5	26.9	42.4	46.5	34.0
Private vehicle, ownership traced	4649	6.4	8.1	9.1	4.6	14.9
Private vehicle, owners unidentified	11200	15.4	11.5	14.3	17.1	20.6
State	3015	4.1	0.8	5.1	5.2	8.8
Widely held, with a blockholder	16052	22.0	24.6	24.6	20.5	19.4
Widely held, no blockholder	8411	11.6	28.1	4.6	6.1	2.4
All firms	72807	100.0	19021	8977	42742	2067
Panel B: Record tiers of discovered controllers and audit coverage						
Candidate discovered records (audited)	47392	7.6	–	–	–	–
verified discovered controllers	165	0.3	–	–	–	–
unverified vendor-sourced controllers	1373	2.9	–	–	–	–
held out (quarantined at baseline)	45854	96.8	–	–	–	–
Records outside the discovery audit	576315	92.4	–	–	–	–
re-weighting TR-known holdings	143599	24.9	–	–	–	–
owner absent, unaudited candidates	35682	6.2	–	–	–	–
owner below 3% floor or ambiguous	250484	43.5	–	–	–	–
firm not matched to a TR issuer	142696	24.8	–	–	–	–
dual-class records (own channel)	3854	0.7	–	–	–	–

Note: Series, bases, and caveats as defined in Section 2.5. Panel A classifies the 72,807 control-chain firms (Table 2, Panel A) by controlling-owner type on the Aminadav–Papaioannou typology; *All (%)* is the 49-market share and the regional columns each region's share, the final row each region's firm count. Panel B grades the owner-level overlay records by their standing in the discovery audit of the 47,392 owner-absent candidates, on the R15 MAIN∪SECONDARY basis. The overlap audit demoted 8 of the 1,381 vendor-sourced records to quarantine, leaving 1,373, the audited total unchanged. The remaining 576,315 overlay records fall outside the audit: 143,599 re-weight holdings TR already carries, 35,682 are further owner-absent candidates, 250,484 name owners below the 3% floor or with an ambiguous match, 142,696 belong to firms unmatched to TR, and 3,854 are dual-class holder records. Panel B percentages are shares of the enclosing total: the 623,707 overlay records (group rows), the 47,392 audited candidates (tier rows), and the 576,315 unaudited records (component rows).

The two audited datasets span the 49 markets and overlap substantially with the co-authors' panel. Within these data, treated firms have economically large control–cash-flow divergences, and the record grading separates verified discoveries from vendor material with a disclosed error rate. The following exhibits report the effects on measured common ownership.

5 Coverage and treatment under the control correction

The control correction replaces each firm's baseline one-share-one-vote control vector with a corrected vector built from ownership-chain tracing and, where they apply, dual-class voting rights. Table 5 reports, for the ten largest markets in the κ panel and for all 49 markets pooled, how many listed firms the correction covers, how many it changes, through which mechanism, and how large the induced change is; the full per-market table is Table 19 in Appendix A. A firm is *covered* when the correction computes a corrected control vector for it; a firm is *treated* when that correction alters its control vector, either because ownership-chain tracing reassigns control away from the direct registered holder or because dual-class voting rights separate votes from cash-flow rights. The mechanism buckets (chains only, dual-class only, and both) are disjoint and sum to the total treated count.

Across the 49 markets, the correction covers 54,488 firms and treats 20,842 of them (38.3%), and in the average market the treated firms account for 50.4% of covered-firm market capitalisation, so treatment is concentrated among larger firms. The control-chain channel accounts for nearly all treated firms. Among all treated firms, 20,292 are chains-only, 338 are dual-class-only, and 212 are treated by both mechanisms. Treatment rates and the composition of mechanisms vary substantially across markets. These counts are cross-sectional and are computed on the demoted permissive variant; the basis, the count of firms the headline series actually treats, and the sample screens that bound the census universe are set out after the table.

Table 5: Coverage and treatment of the control correction: ten largest markets and pooled total

	Firms covered	Treated firms (by mechanism)			Total treated	Treated firms (%)	Treated mkt. cap (%)	Mean $\lvert\gamma-\beta\rvert$
		Chains	Dual-cl.	Both				
China	11979	3505	78	0	3583	29.9	52.5	0.158
United States	7195	2064	0	0	2064	28.7	17.4	0.085
India	6149	2464	3	2	2469	40.2	71.1	0.125
Japan	4024	1222	0	0	1222	30.4	25.1	0.122
Korea	2745	1141	36	63	1240	45.2	86.9	0.147
Canada	2475	426	1	1	428	17.3	12.9	0.124
Taiwan	2122	772	2	0	774	36.5	13.4	0.093
Australia	1763	731	0	0	731	41.5	16.7	0.080
Vietnam	1574	775	1	0	776	49.3	77.7	0.151
United Kingdom	1506	886	0	1	887	58.9	39.8	0.098
All countries	54488	20292	338	212	20842	38.3	50.4	0.144

Note: Series, bases, and caveats are as defined in §2.5. Markets are ordered by firms covered; the ten largest are shown here, with the full 49-market table, complete column definitions, and provenance in Table 19 of Appendix A. *Firms covered* counts distinct listed firms for which the correction computes a corrected control vector; *treated firms* are those whose control vector it alters, from an owner-level wedge $\gamma-\beta$ above a 10^{-6} tolerance, split into chains-only, dual-class-only, and *both*. *Treated mkt. cap (%)* is the treated firms' unweighted-mean share of covered-firm capitalisation across markets, and *mean* $\lvert\gamma-\beta\rvert$ the mean over treated firms of the largest within-firm control-share reallocation. The United States enters through ownership-chain tracing only; its DEF 14A dual-class voting vectors are reported separately in the reinstatement analysis of Section 8 and are excluded from this overlay. Counts are cross-sectional on the demoted permissive variant at reference quarter 2023Q4.

5.0.0.1 Basis of the treated counts.

The covered and treated counts in Table 5 are cross-sectional and are computed on the demoted permissive variant (§2.5). A firm is counted as treated whenever any candidate owner record moves its control vector, including records the headline series subsequently holds out under record grading, so the count of firms the headline series actually treats is smaller. At the 2023Q4 reference quarter the headline series applies a changed control vector to 27,174 of the 37,639 TR-panel firm-quarters (72.2%), against 27,690 (73.6%) under the permissive variant, so record grading reverts 516 firm-quarters – just under 2% of the permissive applied set – to baseline; at 2024Q4 the corresponding counts are 22,024 (75.8% of 29,037 firm-quarters) under the headline series and 22,172 (76.4%) under the permissive variant, with 148 reverted. The headline series therefore reaches nearly the same firms as the permissive variant. Record grading has little effect on the count of firms reached and primarily changes the magnitude of each firm's correction. The median market's $\bar{\kappa}$ falls by 0.37% under the headline series against 26.7% under the permissive variant at 2023Q4 (0.83% versus 38.3% at 2024Q4), with the country-level movements developed in Sections 6 and 10. The applied-under-headline figures are drawn from the R15 netting recompute (*r15_applied_numbers.json*).

5.0.0.2 Scope of the sample screens.

The census covers the firms present in the co-authors' estimation panel, which applies several sample screens documented in their *main_OSOV.tex* besides the multiclass exclusion: depositary receipts are excluded; firm-quarters in which a single holder holds more than 98.5% are dropped; firm-quarters whose ownership ratio exceeds 120% are excluded and those between 100% and 120% are re-weighted; firm-quarters with less than 5% ownership coverage are excluded; and low-coverage countries are dropped entirely. The reinstatement analysis of Section 8 undoes *only* the multiclass screen: firms removed by the other screens remain outside both the baseline and the enlarged panels, so the census and the enlargement pertain only to the multiclass-excluded universe. Two of these screens bear on the direction of the control correction and warrant the co-authors' attention. The single-holder $> 98.5\%$ screen and the $< 5\%$ coverage screen both remove disproportionately concentrated or tightly held firms – the firms whose control the correction most often reassigns – so the surviving panel's measured common ownership is higher than the full listed universe's, and the estimation sample understates the concentrated tail the correction targets. This is a documentation-based observation; the co-authors' screen implementation has not been independently audited.

The correction reaches about two firms in five within each market's listed universe and half of covered capitalisation. The levels and survival sections report the resulting changes in measured common ownership, and the summary-statistics section describes the underlying datasets in detail.

6 Corrected levels of common ownership

After accounting for control chains and dual-class voting rights, the corrected country-average profit weight $\bar{\kappa}$ moves only modestly at the centre of the cross-section while remaining sizeable in the concentrated-ownership tail. The unweighted cross-market average change is −0.7% at the composition-stable 2023 annual mean and −3.1% at the 2024Q4 cross-section, and the median market moves by −0.3% and −0.8% respectively; the dispersion is wide, running from reductions near a sixth of the baseline level in the most concentrated-ownership economies to increases of similar size where the identity correction adds verified controllers together with the common-ownership links they carry. Throughout, the displayed series is the headline series defined in §2.5 (the MAIN∪SECONDARY control assignment combined with the dual-class voting correction); the demoted permissive variant (*combined_trace_DC_R1*) appears only as a labeled robustness leg in the channel-decomposition summary. The attribution exhibit decomposes the movement by record grade. The country panel below reports the levels market by market, the figure that follows ranks the markets by the size of the change, and the closing decomposition splits the correction into its two channels.

6.1 Country panel of $\bar{\kappa}$, baseline vs. corrected

Under the headline series the correction generally lowers measured common ownership, reducing $\bar{\kappa}$ in 28 of the 49 markets at the composition-stable 2023 annual mean and in 33 markets at the terminal 2024Q4 cross-section, raising it in 20 and 14 markets respectively where the net identity correction adds controllers and their links, and leaves Russia and Saudi Arabia unchanged. The size of the change ranges from a reduction of about a sixth of the baseline level to an increase of about a sixth. Table 6 reports the country-average profit weight $\bar{\kappa}$ under the baseline one-share-one-vote assignment and under the correction, at both the composition-stable 2023 annual mean and the terminal 2024Q4 cross-section.

Table 6: Country panel: levels of $\bar{\kappa}$, baseline vs. corrected

	2023 annual mean			2024Q4		
	Baseline	Corrected	Δ%	Baseline	Corrected	Δ%
Australia	0.009	0.009	−6.0	0.035	0.036	4.0
Austria	0.010	0.012	14.3	0.019	0.022	16.7
Belgium	0.015	0.015	0.8	0.023	0.023	−1.3
Brazil	0.009	0.008	−8.9	0.081	0.071	−12.7
Canada	0.005	0.005	−5.3	0.031	0.028	−8.4
Chile	0.019	0.018	−0.9	0.020	0.018	−9.4
China	0.048	0.048	1.2	0.044	0.045	1.2
Colombia	0.014	0.014	−2.5	0.016	0.015	−1.0
Croatia	0.030	0.029	−1.3	0.041	0.040	−0.7
Denmark	0.020	0.020	3.0	0.026	0.026	0.7
Egypt	0.064	0.064	0.0	0.069	0.069	0.3
Finland	0.039	0.036	−6.4	0.044	0.041	−7.6
France	0.007	0.007	2.6	0.013	0.013	0.1
Germany	0.012	0.011	−10.4	0.018	0.016	−9.3

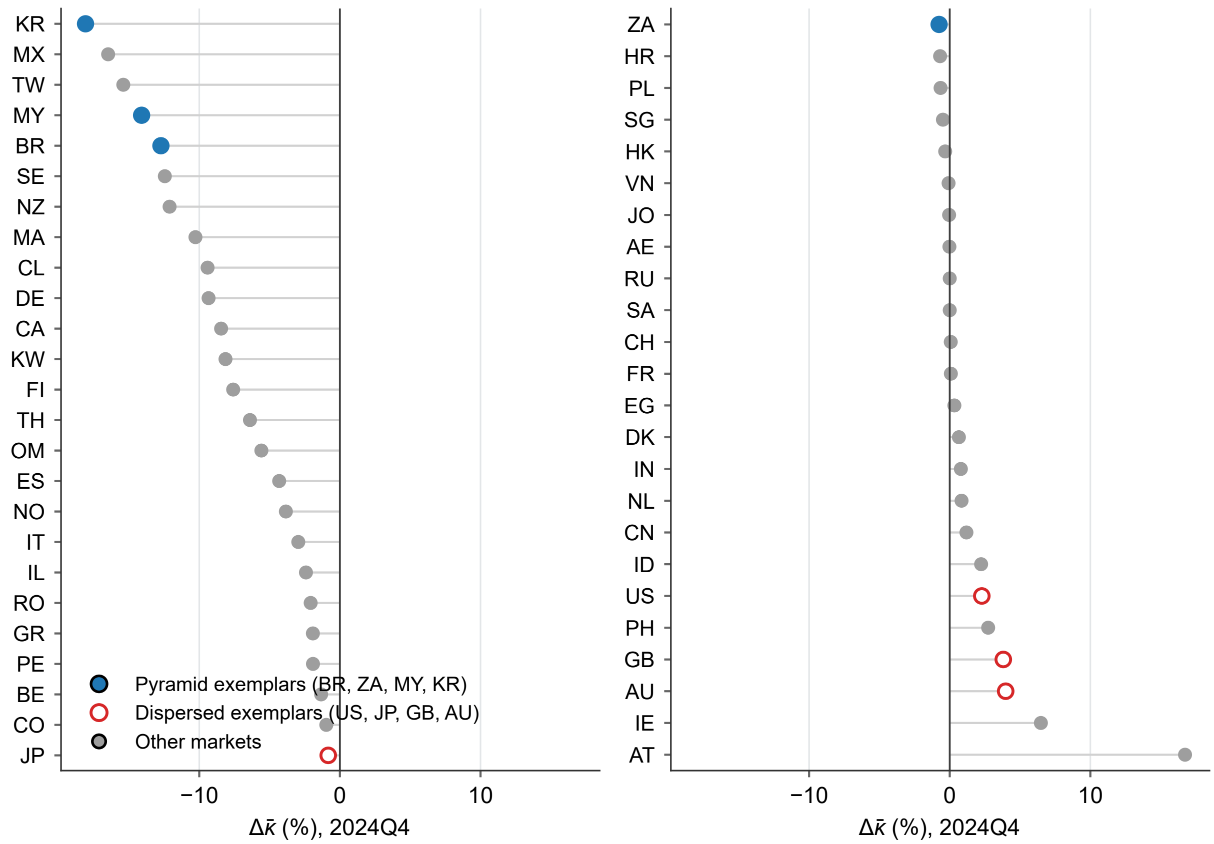
	2023 annual mean			2024Q4		
	Baseline	Corrected	Δ%	Baseline	Corrected	Δ%
Greece	0.004	0.004	−1.1	0.004	0.004	−1.9
Hong Kong	0.007	0.007	−0.3	0.016	0.016	−0.3
India	0.001	0.001	1.6	0.001	0.001	0.8
Indonesia	0.000	0.000	2.5	0.000	0.000	2.2
Ireland	0.128	0.131	2.8	0.203	0.216	6.5
Israel	0.014	0.014	0.6	0.017	0.016	−2.4
Italy	0.006	0.006	2.2	0.032	0.031	−3.0
Japan	0.027	0.027	0.4	0.213	0.212	−0.8
Jordan	0.035	0.034	−0.1	0.026	0.026	−0.0
Korea	0.002	0.002	−19.5	0.002	0.002	−18.1
Kuwait	0.028	0.028	−1.5	0.031	0.029	−8.1
Malaysia	0.003	0.003	−3.5	0.026	0.023	−14.1
Mexico	0.039	0.033	−15.8	0.040	0.033	−16.5
Morocco	0.018	0.024	35.5	0.020	0.018	−10.3
Netherlands	0.044	0.045	2.1	0.070	0.070	0.8
New Zealand	0.040	0.039	−4.8	0.112	0.098	−12.1
Norway	0.017	0.017	1.9	0.021	0.020	−3.8
Oman	0.117	0.111	−4.8	0.117	0.111	−5.6
Peru	0.007	0.007	0.4	0.016	0.016	−1.9
Philippines	0.003	0.003	2.5	0.003	0.003	2.7
Poland	0.004	0.004	−2.4	0.003	0.003	−0.7
Romania	0.018	0.017	−3.5	0.020	0.020	−2.1
Russia	0.010	0.010	0.0	0.009	0.009	0.0
Saudi Arabia	0.026	0.026	−0.3	0.037	0.037	0.0
Singapore	0.002	0.002	3.0	0.009	0.009	−0.5
South Africa	0.124	0.124	−0.0	0.180	0.178	−0.8
Spain	0.007	0.006	−2.7	0.006	0.006	−4.3
Sweden	0.043	0.041	−6.5	0.041	0.036	−12.4
Switzerland	0.074	0.082	11.0	0.096	0.096	0.1
Taiwan	0.007	0.007	−6.9	0.037	0.031	−15.4
Thailand	0.002	0.002	−7.6	0.003	0.003	−6.4
United Arab Emirates	0.032	0.031	−3.5	0.029	0.029	−0.0
United Kingdom	0.039	0.039	−0.6	0.059	0.061	3.8
United States	0.191	0.199	3.9	0.353	0.361	2.3
Vietnam	0.003	0.003	−0.1	0.003	0.003	−0.1
Unweighted mean	0.029	0.029	−0.7	0.048	0.047	−3.1

Note: Series, bases, and caveats are as defined in §2.5. Country-average profit weights $\bar{\kappa}$: *Baseline* is the one-share-one-vote assignment and *Corrected* is the headline series, each shown at the 2023 annual mean (the simple average of the four 2023 quarters) and at 2024Q4, with $\Delta\% = 100 \times (\text{corrected/baseline} - 1)$. Rows are the 49 covered markets, all with full 2023 coverage and present at 2024Q4; the final row is the unweighted cross-country average of each column. The 2024Q4 cross-section carries the roster-attrition caveat of §2.5: in aggregate the covered universe shrinks by about 23% between 2023Q4 and 2024Q4 (from 37,639 to 29,037 firms, with Japan falling by roughly a third in the final quarter alone, −32% from 2024Q3, and per-country rates ranging widely from about −2% to −79%), so the terminal-quarter levels rest on a smaller and probably differently composed roster while the 2023 annual mean is the composition-stable companion.

6.2 Δ% by market

Figure 1 ranks all 49 markets by the change at 2024Q4 and summarises the cross-sectional pattern. The largest reductions fall in the concentrated-ownership economies: Korea, Mexico, Taiwan and Malaysia head the ranking, each losing 14 to 18 percent of baseline common ownership. Three of the four pyramid exemplars (Korea, Malaysia, Brazil) sit among the five deepest reductions, while the fourth, South Africa, falls to the middle of the ranking (26th), its large permissive reduction resting on discovered-controller records the composed headline does not carry. The dispersed-ownership exemplars cluster at the opposite end, where the correction raises measured common ownership: the United States rises 2.3% (44th of 49), the United Kingdom 3.8% and Australia 4.0%, because for these markets the net effect of resolving controller identities is to add verified controllers and the common-ownership links they carry, with no corresponding removal of control from dispersed holders. The change is therefore largest where the proportional-control assumption is least defensible and can be positive where it is most defensible; the ordering remains continuous, as the note details.

Figure 1: Corrected-vs-baseline $\Delta\bar{\kappa}$ (%) by market, 2024Q4



Note: Series, bases, and caveats are as defined in §2.5. Percent change in the country-average $\bar{\kappa}$ from the baseline one-share-one-vote assignment to the headline series at 2024Q4, all 49 markets sorted from the largest reduction. Filled markers are the pyramid/concentrated-control exemplars (Brazil, South Africa, Malaysia, Korea), open markers the dispersed-ownership exemplars (United States, Japan, United Kingdom, Australia), and gray markers the remaining markets; these exemplar groups are illustrative anchors on a continuous ordering without defining a taxonomy, while a formal regime classification remains a pending co-author decision. The 2024Q4 basis carries the roster-attrition caveat of §2.5; at the 2023 annual mean the cross-market average change is −0.7% against −3.1% at 2024Q4, and the country ordering is only partly preserved across the two bases (rank correlation 0.65, largest rank shift 41 places, 18 markets showing a larger reduction at the 2023 mean than at 2024Q4).

6.3 Decomposition by control channel

The control correction combines two data layers that differ in construction. Ownership-chain tracing reassigns control from registered holders to ultimate owners through the cross-jurisdiction control engine, while the dual-class layer separates votes from cash-flow rights using per-firm adjudicated voting terms read from statutes and primary disclosures. Because a reader may weigh the two kinds of evidence differently, the decomposition reports each layer's effect on average common ownership separately and alongside the combined effect. The composed correction satisfies the additive identity $\Delta_{\text{combined}} = \Delta_{\text{chains}} + \Delta_{\text{dual-class}} + \Delta_{\text{interaction}}$ by construction. A reader who accepts only the control-chain evidence can read the chains channel, while one who accepts only the dual-class evidence can read the dual-class channel. The combined channel recovers the headline correction. The interaction channel records the difference between the combined effect and the two single-channel effects when added. Table 7 summarises the channels across the 49 markets at 2024Q4; the per-market split is reported in Table 21 of Appendix A.

Across the 49 markets the control-chain channel carries almost the entire correction. Its median effect at 2024Q4 is a reduction in average $\bar{\kappa}$ of 0.0002, about 0.8 percent of baseline common ownership in the median market, and it is active in 47 of the 49 markets; the dual-class channel is active in 8 markets with an across-market mean effect of −0.0003. The interaction is negligible in most markets, with a median of zero, and exceeds 10^{-4} in only three markets (Brazil, Sweden, Germany), reaching at most 0.0046 in Brazil, so the combined correction is close to the sum of its two parts everywhere. At the composition-stable 2023 annual mean the two channels are smaller again. The control-chain channel is the larger of the two in 47 of the 49 markets and the dual-class channel is larger only in Brazil and Korea; the interaction reaches at most 0.0003 (in Sweden), and the standalone dual-class channel never exceeds 0.0013 (in Sweden and Austria). Beneath these headline channels the summary table reports the demoted permissive chains leg together with the other robustness legs. The permissive chains channel moves the median market by −0.0091 in $\bar{\kappa}$ (−38.2 percent of baseline), some sixty times the netted MAIN∪SECONDARY channel. Accordingly, the demoted permissive variant is no longer the display. For the United States, the separation shows that the correction in this integration vintage operates through ownership-chain tracing only. Its baseline average $\bar{\kappa}$ of 0.3525 rises by 0.0080 to 0.3606 under the netted headline, an effect entirely attributable to the chains channel with no dual-class contribution. The reinstatement section reports the United States dual-class evidence in its panel-enlargement exhibit.

Table 7: Channel movements in average $\bar{\kappa}$ across the 49 markets (2024Q4)

	Change in $\bar{\kappa}$		% of baseline		Markets active
	Median	Mean	Median	Mean	
Chains (MAIN+SEC)	−0.0002	−0.0007	−0.8	−2.1	47
Dual-class (R1)	0.0000	−0.0003	0.0	−1.2	8
Interaction	0.0000	0.0001	0.0	0.2	8
Combined	−0.0002	−0.0009	−0.8	−3.1	47
Chains (permissive)	−0.0091	−0.0174	−38.2	−34.7	47
Chains (weak-link)	−0.0087	−0.0173	–	–	47
Chains (strict/MAIN)	0.0001	0.0015	–	–	47
Dual-class (R2)	0.0000	−0.0001	–	–	7

Note: Series, bases, and caveats are as defined in §2.5. The table summarises across the 49 markets the per-market channel changes reported in Table 21 of Appendix A. Each row is a correction channel or robustness leg; the columns give the median and mean change in average $\bar{\kappa}$ across markets, the same two statistics as a percentage of baseline, and the count of markets moving $\bar{\kappa}$ by more than 10^{-6}. The four rows above the gap are the composed-headline channels (the control-chain channel under the MAIN∪SECONDARY netting, the dual-class channel under the R1 denominator convention, their interaction, and the combined headline); the four below are robustness legs, of which *Chains (permissive)* is the demoted series retaining every discovered-controller candidate record (the chain leg of *combined_trace_DC_R1*), moving the median market by −0.0091 in $\bar{\kappa}$ (−38.2 percent of baseline), some sixty times the netted headline channel, with *Chains (weak-link)*, *Chains (strict/MAIN)*, and *Dual-class (R2)* the remaining legs. Percentage columns are shown only for the headline channels and the permissive leg; the interaction is small throughout, reaching at most 0.0046, in Brazil.

Under the headline series the correction moves the median market by less than one percent and the average market by a few percent, operating through the control-chain channel in all but a handful of dual-class markets. The reduction is largest where control is most concentrated, while measured common ownership rises where identities resolve into additional verified controllers. The demoted permissive variant, which moved the median market by about −38%, appears only as a labeled robustness leg; the attribution exhibit reports the movement attributable to each grade of discovered-controller record, and the choice among the series is a pending co-author decision. The survival exhibit reports the corresponding time-series results.

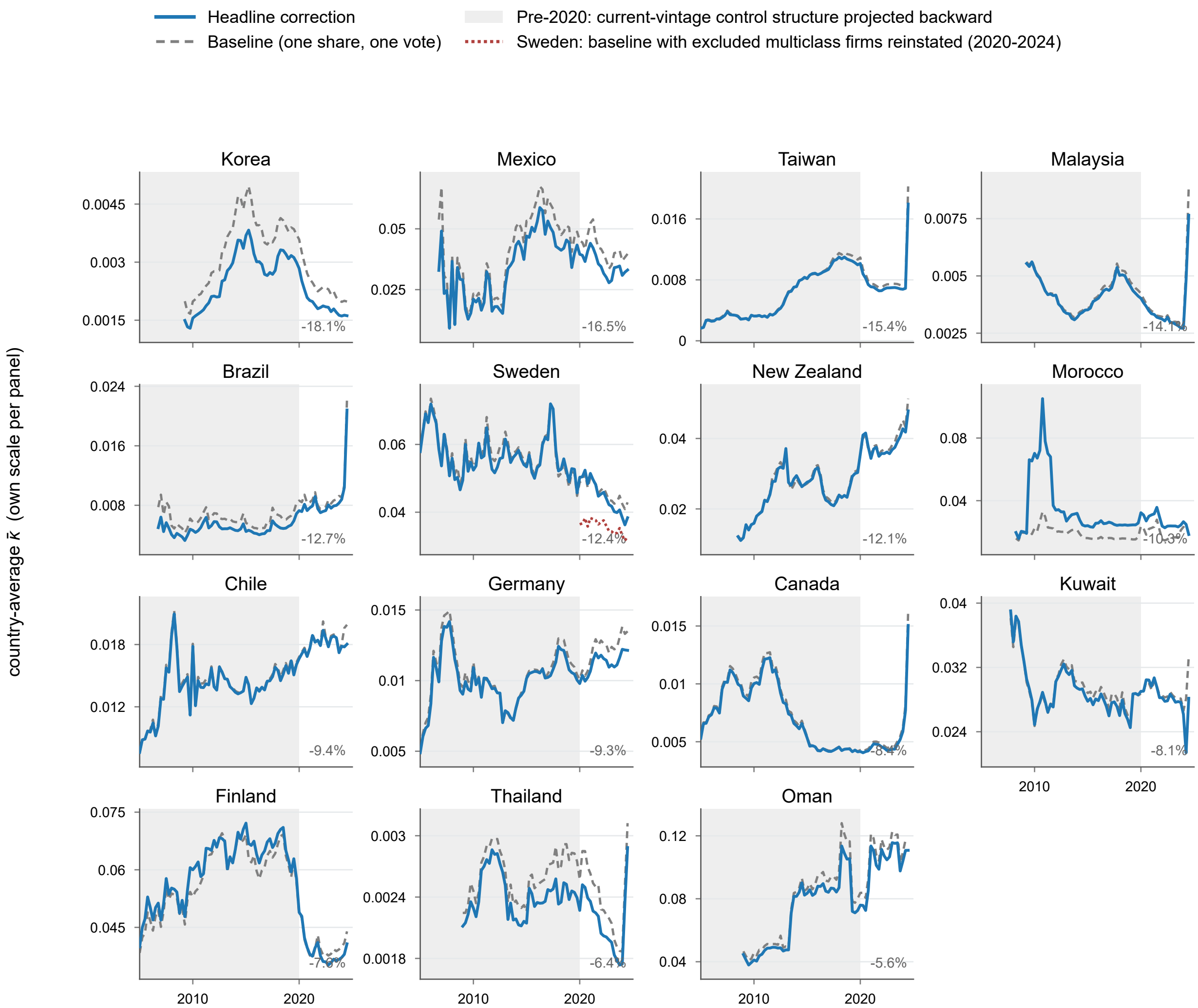
7 Growth of common ownership under the control correction

7.1 Time-series results

The levels exhibits show that the control correction changes measured common ownership modestly and heterogeneously, lowering it at the all-market median while raising it in the United States. The time-series analysis finds that the two-decade rise in common ownership survives the correction across the tabulated markets. Figures 2–4 plot the quarterly country-average $\bar{\kappa}$ over 2005–2024 for each of the 49 jurisdictions, one panel per jurisdiction, under the baseline one-share-one-vote assignment and under the headline series defined in Section 2.5. The panels are grouped across three figures by the sign and size of the 2024Q4 headline correction: the markets where the correction lowers $\bar{\kappa}$ substantially (Figure 2), those where it barely moves $\bar{\kappa}$ (Figure 3), and those where it raises $\bar{\kappa}$ substantially, led by the United States (Figure 4). For the two markets whose excluded multiclass firms have been reinstated (Section 8), the panels also draw the reinstated series over its 2020–2024 build window as a dotted line: with the excluded firms restored, the United States level sits roughly a third below the in-panel corrected line, so the in-panel rise must be read jointly with the exclusion bias that Section 8 measures. Across the groups the corrected trajectory tracks the shape of the baseline path and preserves its trend. The United States correction holds the corrected line above the baseline throughout on the shipped universe, while the pyramidal and family-controlled markets that take the largest negative corrections preserve the direction of their baseline path.

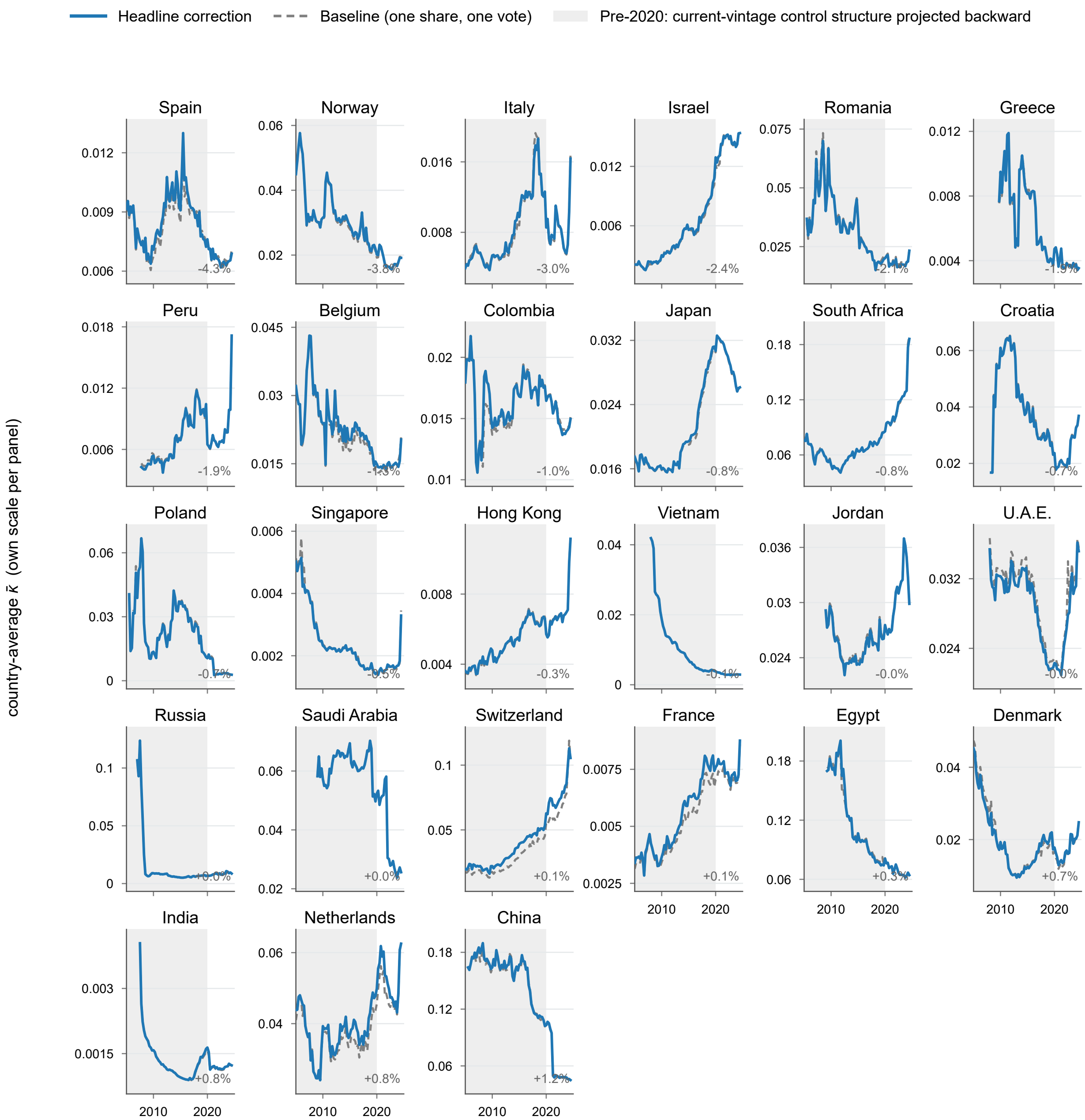
Table 8 reports the corresponding growth estimates. The correction leaves the rise substantially intact wherever the baseline rise was real. The United States grows by 401% under the correction against 425% at baseline (a −24pp difference) while remaining above the baseline in level; the pyramid group's rise steepens, from 130% to 135%; and Japan's rise is preserved (+64.7% against +63.7%). Where the baseline rise was small the correction moves it only modestly and never reverses it: Korea's +1.9% becomes +4.7%, Australia's +9.4% becomes +3.0%, and the dispersed-group average of +5.4% becomes +4.0%. The United Kingdom's and Malaysia's baselines themselves fell over the window, so neither had a rise to preserve. These trajectories depart from those of the demoted permissive variant (*combined_trace_DC_R1*, the former shipped estimator), which retained the discovered-controller records the roster could not corroborate: under that variant the same modest rises turned into declines (Korea −6.1%, Australia −15.0%, the dispersed group −8.2%). The attribution exhibit decomposes the movement by record grade and traces those reversals to the uncorroborated records excluded from the headline series displayed here.

Figure 2: Country-average $\bar{\kappa}$ by jurisdiction: markets where the control correction lowers common ownership, 2005–2024



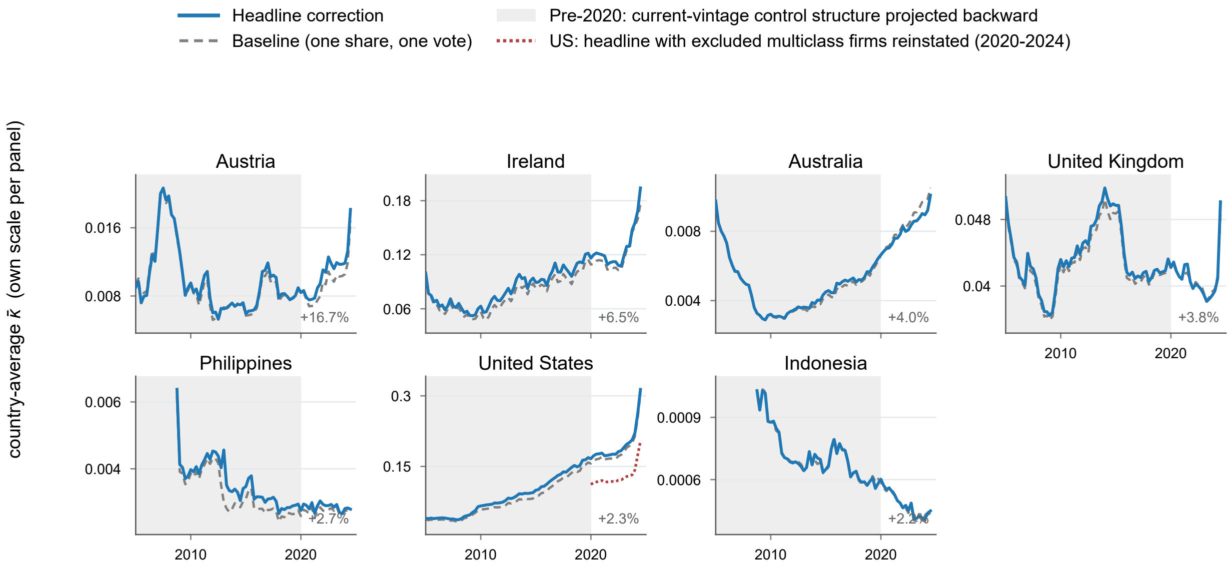
Note: Quarterly country-average $\bar{\kappa}$ under the baseline one-share-one-vote assignment (dashed grey) and the headline series (solid), one panel per jurisdiction on its own vertical scale, for the markets whose 2024Q4 headline correction is below −5% (annotated in each panel, ordered from most negative). The dotted line in the Sweden panel is the reinstated series of Section 8 (the baseline with Sweden's excluded multiclass firms restored, the cross-firm analog build), drawn over its 2020–2024 build window. Construction, the pre-2020 shaded region, the grouping cutoffs, and the series not drawn are described in the note to Figure 4.

Figure 3: Country-average $\bar{\kappa}$ by jurisdiction: markets where the control correction barely moves common ownership, 2005–2024



Note: Quarterly country-average $\bar{\kappa}$ under the baseline one-share-one-vote assignment (dashed grey) and the headline series (solid), one panel per jurisdiction on its own vertical scale, for the markets whose 2024Q4 headline correction lies between −5% and +2% (annotated in each panel, ordered from most negative to most positive). Construction, the pre-2020 shaded region, the grouping cutoffs, and the series not drawn are described in the note to Figure 4.

Figure 4: Country-average $\bar{\kappa}$ by jurisdiction: markets where the control correction raises common ownership, 2005–2024



Note: Series, bases, and caveats as defined in Section 2.5. Quarterly country-average $\bar{\kappa}$ by jurisdiction under the baseline one-share-one-vote assignment (dashed grey) and the headline series (solid), each panel on its own vertical scale (levels span orders of magnitude), which limits cross-panel comparisons to shape. The 49 jurisdictions are split across the three figures by the 2024Q4 headline correction $100(\bar{\kappa}^{\text{headline}}/\bar{\kappa}^{\text{baseline}} - 1)$, the level-correction magnitude reported in the levels exhibit: above +2% here, between −5% and +2% in Figure 3, and below −5% in Figure 2, the two cutoffs sitting at natural gaps in the sorted distribution; each panel is annotated with its correction. The shaded region (2005–2019) applies the 2020–24 control-structure vintage backward, anchoring the corrected series in the contemporaneous 2020–24 window; the overfull-issuer gate that reverts incompatible issuer-quarters to baseline fires on 6 issuers at 2024Q4 but on the order of 479 per quarter by 2019. Trajectories run 2005Q1–2024Q3; the terminal quarter (2024Q4) is excluded for the roster-attrition reason in the note to Table 8 but still defines group membership, matching the level exhibits. The dotted line in the United States panel is the enlarged-universe series of Section 8 (the headline correction with the excluded multiclass firms reinstated under the certified DEF 14A vectors), drawn over its 2020–2024 build window; the solid and dashed lines in every panel are otherwise computed on the shipped universe, which excludes the reinstated firms. The control-chain-only series, the loyalty-restored satellite, and the earlier group-average lines are omitted. The control-chain-only line coincides with the headline outside eight dual-class-active markets; the growth table reports the channel split for those markets and retains the country groups.

Table 8: Growth of $\bar{\kappa}$: baseline vs. corrected, by country group

	Base year	Baseline (OSOV)			Corrected			Δ growth (pp)
		Base	2023	Growth (%)	Base	2023	Growth (%)	
United States	2005	0.0364	0.1913	425.3	0.0397	0.1987	401.1	−24.1
Pyramid group	2010	0.0150	0.0345	129.7	0.0146	0.0342	134.9	5.1
Brazil	2007	0.0081	0.0087	7.6	0.0054	0.0080	48.1	40.5
South Africa	2005	0.0766	0.1243	62.2	0.0766	0.1242	62.3	0.1
Malaysia	2010	0.0050	0.0030	−41.0	0.0050	0.0028	−42.5	−1.5
Korea	2010	0.0021	0.0021	1.9	0.0016	0.0017	4.7	2.8
Dispersed group	2005	0.0239	0.0252	5.4	0.0240	0.0249	4.0	−1.4
United Kingdom	2005	0.0464	0.0388	−16.3	0.0468	0.0386	−17.5	−1.2
Japan	2005	0.0167	0.0274	63.7	0.0167	0.0275	64.7	1.0
Australia	2005	0.0085	0.0093	9.4	0.0085	0.0087	3.0	−6.4

Note: Series, bases, and caveats as defined in Section 2.5. Annual means of the quarterly country-average $\bar{\kappa}$. "Base" is each row's first fully covered calendar year (all four quarters present; Base-year column); for a group it is the first year *every* member is fully covered (2010 for the pyramid group, since Malaysia and Korea enter in 2009Q2). Growth is the percent change in the annual mean from that base year to 2023; Δ growth is corrected minus baseline, in percentage points; 2024 is excluded for the roster-attrition reason given in Section 2.5. The rise survives the correction in every group: the United States grows 401% against a 425% baseline while remaining above the baseline in all 80 panel quarters (by up to 0.014), the pyramid group's rise steepens (135% against 130%), and the dispersed group's is modestly trimmed (4.0% against 5.4%); where the baseline rise was small the correction moves it without reversing it (Korea +1.9% → +4.7%, Australia +9.4% → +3.0%), while the United Kingdom's and Malaysia's baselines themselves fell over the window. The pre-2020 dual-class treatment matters only for the dual-class-active markets, where the control-chain channel alone gives Brazil +13.0% (versus +48.1%), the pyramid group +130.1% (versus +134.9%), and Korea +2.8% (versus +4.7%), while every other row moves by at most 0.6pp. Where the base $\bar{\kappa}$ is near zero, percent growth is sensitive to the base level, so Δ growth is reported alongside: for Korea the base and 2023 means are $\bar{\kappa} = 0.00209 \rightarrow 0.00213$ (baseline) and $0.00164 \rightarrow 0.00171$ (corrected).

The correction therefore lowers the level of common ownership in most markets and raises it in the United States, while preserving the two-decade rise in the United States, the pyramid group, and the dispersed group; no baseline rise becomes a decline. These trajectories come from the headline series, which excludes the uncorroborated discovered-controller records retained by the demoted permissive variant. The attribution exhibit reports the movement attributable to each record grade, and the remaining exhibits describe the excluded firms, the US voting wedges, and the attribution of the correction across record grades.

8 Effects of reinstating excluded multiclass firms

The multiclass exclusion is a first-order limitation of the shipped panel: it removes exactly the firms whose share structures separate control from ownership, and in doing so it biases the level of $\bar{\kappa}$ and hides real common-ownership linkage among the firms that remain. This section measures that bias. The measurement requires both universes, because the bias is the difference between the shipped panel and the panel with the excluded firms restored: the central exhibits therefore carry the shipped universe, and this section reports what changes when the excluded firms enter, decomposing the movement into a mechanical dilution and a genuine linkage gain. The reinstated control vectors carry the same per-firm adjudication as the rest of the dual-class data (the certified DEF 14A derivation for the United States and the register build for Sweden), so the separation reflects the structure of the comparison and the current reach of the data, never an evidentiary distinction. The exclusion itself is panel-wide: the co-authors' construction removes 3,620 multiclass issuers across 41 markets, censused market by market below and in Table 22. The reinstatement append is so far executed for the United States and Sweden only, which hold 1,819 of the excluded firms; extending the append to the remaining markets is a stated limitation of this draft and a priority for the next stage.

The co-authors' shipped κ panel excludes multiclass (dual-class) firms, which reinstatement adds back at two scopes. In the *enlarged panel*, the reinstated firms enter as new focal firms. The *incumbent* scope covers firms that were already present and whose own profit weights move once the reinstated firms join. Each reinstated firm enters carrying its corrected dual-class control vector where one is active for that firm-quarter; a reinstated firm without an active corrected vector still enters the panel as a dilution-only node, affecting the averaging denominator but contributing no new common-ownership links. At the 2024Q4 terminal quarter, 450 of the 946 United States reinstated firms carry an active corrected vector and the remaining 496 enter as dilution-only nodes, while all 79 Swedish reinstated firms carry vectors. Because the number of active corrected vectors rises steeply near the sample edge, the headline incumbent decomposition below is reported as the 2023 annual mean, with the 2024Q4 quarter shown as a labeled sample-edge variant.

At the enlarged-panel scope, the average falls sharply once the excluded firms enter. The United States 2023-mean $\bar{\kappa}$ drops from 0.191 to 0.124 (−35.1%), and the drop at the 2024Q4 edge is similar in proportion (−36.1%). At the incumbent scope, an incumbent firm f carries profit weight $\bar{\kappa}_f = (n-1)^{-1}\sum_{g\neq f}\kappa_{fg}$ over a panel of n firms, and appending the reinstated firms changes it by an amount that decomposes exactly into a mechanical dilution and a linkage term,

$$\Delta\bar{\kappa}_f = \underbrace{\left(\tfrac{1}{n_{\text{tot}}-1} - \tfrac{1}{n_{\text{corr}}-1}\right) \sum_{g\in\text{inc},\, g\neq f} \kappa_{fg}}_{\text{dilution}} + \underbrace{\frac{1}{n_{\text{tot}}-1} \sum_{r\in\text{reinst}} \kappa_{fr}}_{\text{linkage}},$$

where n_{corr} and n_{tot} are the incumbent-only and enlarged panel sizes. The dilution term is mechanical: each incumbent's $\bar{\kappa}_f$ averages its pairwise weights over the other panel firms, so enlarging the panel spreads the incumbent's unchanged link sum over more peers and lowers the average even if the entrants brought no common-ownership links at all. The linkage term is the incumbent-to-reinstated link sum ($inc\_to\_new\_sum_f = \sum_{r\in\text{reinst}} \kappa_{fr}$) divided by $n_{\text{tot}} - 1$, and it is non-negative because every $\kappa_{fr} \geq 0$. Averaged over incumbents, mechanical dilution accounts for -0.0377 of the -0.0372 United States total at the 2023 mean. The linkage component measures the common-ownership connection incumbents acquire through the reinstated firms' owners and is positive in every one of the twenty quarters in both countries (+0.0005 for the United States and +0.0009 for Sweden at the 2023 mean). Excluding the multiclass firms therefore overstates the panel-average level of $\bar{\kappa}$ while suppressing a small but systematic amount of real linkage among the firms that remain.

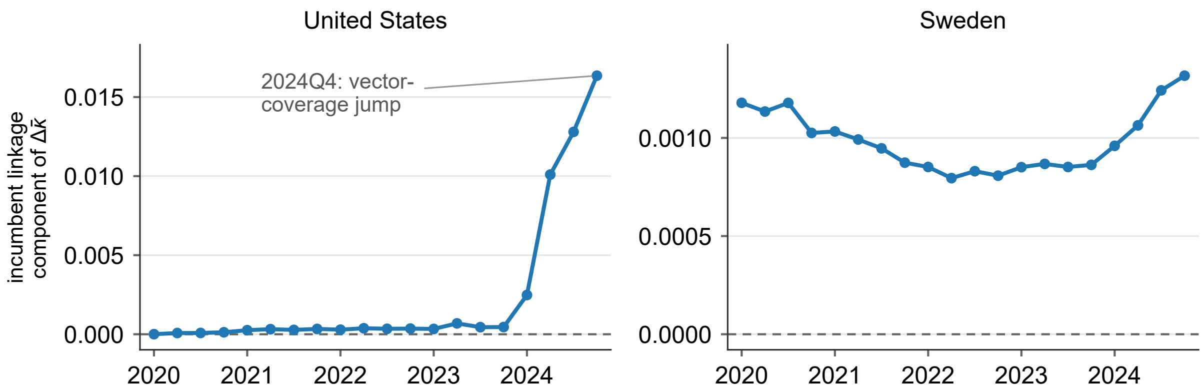
Table 9: Reinstatement of excluded multiclass firms: two scopes

	United States	Sweden
Reinstated multiclass firms (N)	946	79
Reinstated-firm vector coverage (2024Q4)		
Carrying corrected control vector (N)	450	79
Entering as dilution-only node (N)	496	0
Enlarged-panel scope		
Shipped-panel firms (N, 2024Q4)	3,087	432
Enlarged-panel firms (N, 2024Q4)	4,033	511
Shipped-panel $\bar{\kappa}$ (2023 mean)	0.191	0.043
Enlarged-panel $\bar{\kappa}$ (2023 mean)	0.124	0.034
$\Delta\bar{\kappa}$ level (2023 mean)	−0.067	−0.009
$\Delta\bar{\kappa}$ % (2023 mean)	−35.1	−21.1
Shipped-panel $\bar{\kappa}$ (2024Q4)	0.353	0.041
Enlarged-panel $\bar{\kappa}$ (2024Q4)	0.225	0.031
$\Delta\bar{\kappa}$ level (2024Q4)	−0.127	−0.010
$\Delta\bar{\kappa}$ % (2024Q4)	−36.1	−23.8
Incumbent scope: 2023 mean (headline)		
Total movement	−0.0372	−0.0048
Dilution (mechanical)	−0.0377	−0.0057
Linkage (genuine)	0.0005	0.0009
Incumbent scope: 2024Q4 (sample-edge variant)		
Total movement	−0.0664	−0.0050
Dilution (mechanical)	−0.0827	−0.0063
Linkage (genuine)	0.0164	0.0013
Linkage: quarters positive	20/20	20/20

Note: Series, bases, and caveats as defined in Section 2.5. Reinstatement adds the excluded multiclass firms back at two scopes. *Enlarged-panel scope* lets them enter as new focal firms and compares the panel-average profit weight $\bar{\kappa}$ before (shipped panel) and after (enlarged panel); *incumbent scope* restricts attention to firms already in the shipped panel and reports the equal-weighted change in their own $\bar{\kappa}$, decomposed as total = dilution + linkage by the displayed identity above, with *linkage* positive in all twenty quarters in both countries. Coverage (2024Q4): 450 of the 946 United States reinstated firms carry an active corrected control vector and 496 enter as dilution-only nodes, whereas all 79 Swedish firms carry vectors. The decomposition is reported as the 2023 annual mean (headline) and, separately, at the 2024Q4 sample edge, whose United States linkage is inflated by the terminal-quarter jump in coverage (Figure 5) and is shown only as a variant. The United States reinstates 946 representative CIK nodes bridged from its 1,603 excluded multiclass issuers and Sweden 79 of 216; levels are shown to three decimals, the decomposition to four. The United States series is the certified temporal build *b1-us-append-v3.1* and Sweden the *b1-sweden-v2* cross-firm analog.

Figure 5 traces the linkage component quarter by quarter. It remains positive throughout the five-year window, including before the terminal-quarter coverage jump that increases its United States magnitude at the edge.

Figure 5: Incumbent linkage over time, 2020Q1–2024Q4



Note: Series, bases, and caveats as defined in Section 2.5. The linkage component of the incumbent $\bar{\kappa}$ movement, by quarter, for the United States (left) and Sweden (right), each panel on its own vertical scale with a dashed zero line. Linkage is the non-mechanical component of the incumbent decomposition in Table 9, measuring the common-ownership connection incumbents gain through the reinstated firms' owners; it is positive in every one of the twenty quarters in both countries, so reinstating the excluded firms adds common-ownership linkage in addition to diluting denominators. The United States terminal quarter (2024Q4) shows a pronounced increase driven mainly by vector coverage, which rises from 169 reinstated firms carrying an active corrected control vector in 2023Q4 to 450 in 2024Q4. The resulting 35.9× growth in mean linkage over that step factors exactly as a 25.3× coverage effect times a 1.42× incumbent-roster factor, and the always-positive sign of the linkage does not depend on that quarter.

8.1 Panel-enlarged variant: the headline series composed with the reinstated United States firms

The reinstatement above appends the excluded firms to the co-authors' uncorrected panel. A companion series instead composes the reinstatement with the headline series, so that the in-panel United States firms carry their corrected MAIN ∪ SECONDARY control weights while the 946 bridged excluded firms are appended; we label it *mainsec_DC_R1_plusUSreinst*. Because it enlarges the estimation panel beyond the co-authors' current screen, it is reported as a labeled variant. Adoption of the enlarged sample remains a sample-definition decision listed among the pending items in the concluding section.

The interaction between appending dual-class firms and a United States correction that the census records as chains-only depends on the two roles a firm plays. Panel membership governs only its role as a *portfolio* firm, a focal row in the average-over-other-firms that defines $\bar{\kappa}$, whereas its stakes *in* other firms sit on the owner side of the data whether or not the firm is itself in the panel. The in-panel control-chains correction already resolves the corporate-conduit case in which one listed firm holds another. Under this distinction, the in-panel United States headline is invariant to the dual-class leg because the co-authors exclude United States multiclass firms from the panel, leaving no in-panel wedge to correct and routing the whole United States correction through the control-chains channel. Reinstating the excluded firms also adds *new* focal-firm pairs without rewiring the incumbent-incumbent block of the κ matrix, so the incumbent movement decomposes into a mechanical dilution from the weakly linked new pairs and a positive linkage from the common ownership incumbents genuinely share with the reinstated firms.

The two corrections do not combine by simple addition because the headline series replaces in-panel firms' control vectors, changing those firms' links to the reinstated firms. The enlarged panel is therefore recomputed over the corrected firm set and cannot be obtained by summing the two effects separately. At 2024Q4, adding the reinstatement's panel-enlargement change to the headline level would imply $\bar{\kappa} = 0.2331$, while the exact recompute gives 0.2301; the -0.0030 difference is the interaction the recompute captures, an order of magnitude smaller than under the demoted permissive variant and of the opposite sign. The recompute reuses the same engines, reproducing the R15 MAIN $\cup$ SECONDARY netting basis to machine precision.

Table 10 and Figure 6 report the United States series under three lenses. At 2024Q4 the panel-average $\bar{\kappa}$ rises from the baseline 0.353 to 0.361 under the headline in-panel correction (a +2.3% move on 2,518 treated firms, reversing the sign of the demoted permissive variant), then falls to 0.230 once the excluded firms are appended (-0.130, or -36.2% relative to the corrected level). On the incumbent firms alone, measured against the headline-series panel, the variant moves the average $\bar{\kappa}$ by -0.0678 at 2024Q4, decomposing into a dilution of -0.0846 and a linkage of $+0.0168$; at the 2023 annual mean the movement is $-0.0386 = -0.0391 + 0.0005$. The linkage is positive in all twenty quarters, the same pattern as the baseline reinstatement, so enlarging the panel adds real common-ownership linkage while the reinstated firms' low profit weights lower the panel level.

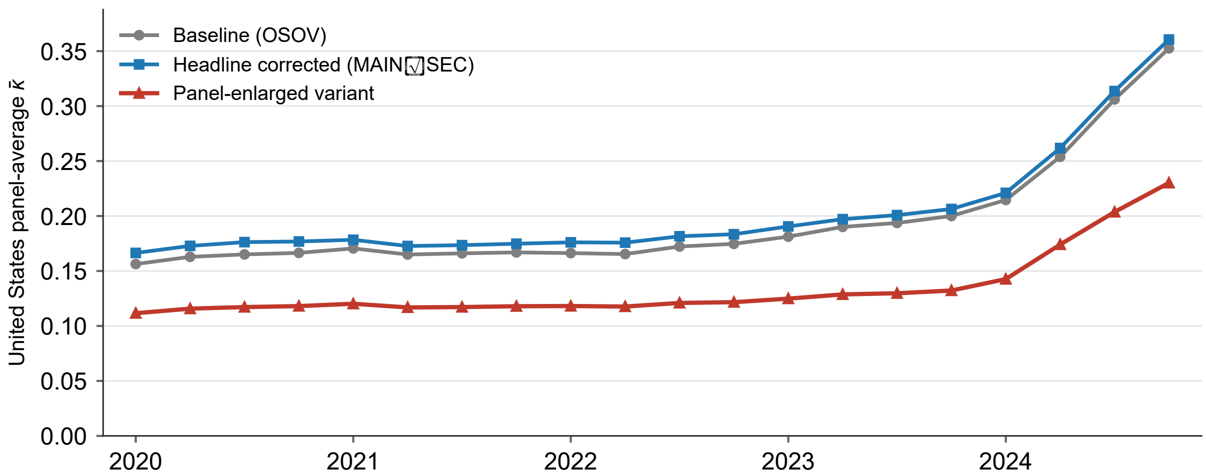
The ownership underlying the positive linkage predates the enlargement. Every incumbent-to-reinstated link was already carried in the co-authors' Thomson–Reuters ownership data; the multiclass screen had deleted the *pairs* those links populate, and reinstatement restores them. A cross-link forms only through an owner the two firms share, and the shared owners are the ordinary institutional cross-holders (the index and 13F managers that hold both the incumbents and the reinstated, index-held dual-class mega-caps). Decomposing the 2024Q4 incumbent-to-reinstated link mass by owner provenance confirms this: 100.000% of the mass flows through owners already present in the Thomson–Reuters data and 0.0000% through overlay-introduced controllers, and the 123 reinstated firms whose disclosed holders are exclusively overlay-introduced controllers carry zero incumbent links. The discovered multi-firm controllers of the Gates–Cascade type form a genuine, second-order additive layer within the in-panel control-chains channel, outside the enlargement channel, which introduces no new ownership.

Table 10: United States panel-enlarged variant: headline series composed with reinstated excluded firms

	2024Q4	2023 mean
Panel-average $\bar{\kappa}$		
Baseline (co-authors' OSOV)	0.353	0.191
Headline corrected (MAIN∪SEC)	0.361	0.199
Panel-enlarged variant	0.230	0.129
Incumbent movement (variant vs. headline-corrected panel)		
Total movement	−0.0678	−0.0386
Dilution (mechanical)	−0.0846	−0.0391
Linkage (genuine)	0.0168	0.0005
Reinstated-firm vector coverage (2024Q4)		
Carrying corrected control vector (N)	450	
Entering as dilution-only node (N)	496	
Linkage: quarters positive	20/20	

Note: Series, bases, and caveats as defined in Section 2.5. The labeled variant *mainsec_DC_R1_plusUSreinst* composes the headline in-panel control correction (the MAIN ∪ SECONDARY estimator on the R15 netting basis, applied to in-panel United States firms) with the reinstatement of the 946 excluded multiclass firms, recomputed jointly over the enlarged, corrected firm set. *Panel-average $\bar{\kappa}$* reports the United States average profit weight under three lenses: the co-authors' baseline (OSOV), the headline correction of in-panel firms, and the panel-enlarged variant. *Incumbent movement* reports the equal-weighted change in the in-panel firms' own $\bar{\kappa}$ from appending the reinstated firms, measured against the headline-series panel and decomposed as total = dilution + linkage by the displayed identity above, with linkage positive in all twenty quarters. The two corrections do not compose additively, because correcting in-panel weights changes those firms' links to the reinstated firms; at 2024Q4 the exact recompute (0.2301) differs from the naive additive value (0.2331) by −0.0030. Levels are shown to three decimals and the decomposition to four, and the corrected in-panel average reproduces the R15 MAIN ∪ SECONDARY netting basis to machine precision. All numbers on this variant are non-citable pending the R3 cross-model audit.

Figure 6: United States panel-average $\bar{\kappa}$ under the three lenses, 2020Q1–2024Q4



Note: Series, bases, and caveats as defined in Section 2.5. The United States panel-average profit weight $\bar{\kappa}$ by quarter under three lenses: the co-authors' baseline (OSOV), the headline correction of in-panel firms (the MAIN ∪ SECONDARY estimator), and the panel-enlarged variant *mainsec_DC_R1_plusUSreinst* that additionally appends the 946 reinstated excluded firms. The in-panel correction shifts the level slightly upward in every quarter; enlarging the panel lowers it below the baseline. The gap between the corrected and enlarged series widens toward the sample edge as active-vector coverage of the reinstated firms rises, whereas the baseline-to-corrected gap reflects the in-panel correction alone and does not depend on reinstatement coverage. All numbers on this variant are non-citable pending the R3 cross-model audit.

Table 23 in Appendix A restates the two reinstatements in the channel-decomposition format of the levels section, placing the panel-enlargement channel beside the in-panel channels on the same footing. It reports the United States move from a reference $\bar{\kappa}$ of 0.3606 to an enlarged 0.2301 (a Δ of -0.1304, of which -0.0678 is incumbent movement decomposing into -0.0846 dilution and $+0.0168$ linkage) and the corresponding Sweden move from 0.0406 to 0.0309; because the reinstated firms are dual-class-routed by construction, their ownership-chain layer is zero by construction rather than by measurement.

8.2 The scope of the exclusion across markets

The reinstatement results in Table 9 cover the two markets whose excluded multiclass firms have been derived, audited, and appended. Table 11 reports those two markets, and Appendix A (Table 22) gives the full census of the exclusion across all markets, thereby quantifying the remaining work. The co-authors' construction removes 3,620 multiclass issuers across 41 markets before the panel is built. Of these, 306 reappear in the 2020–2024 panel through per-security overlap; 245 of the 306 carry a mapped in-panel overlay correction vector, while the remaining 61 are present in the panel but not yet corrected by the overlay. The complementary 3,314 firms form the reinstatement universe. The United States and Sweden account for 1,819 of the excluded firms and for all 1,025 firms reinstated under a prior certified build to date, so the reinstated share of the reinstatement universe is concentrated in the two markets where the disclosure spine was built first. The *reinstated (prior)* column of Table 22 records that earlier build and predates the market-wave and variant appends; the current position of every excluded firm, across all reinstatement routes, is given in Table 12 of the disposition accounting below.

Table 11: Reinstated markets: excluded multiclass firms and reinstatement status (excerpt of the full census, Table 22)

Market	Excluded (N)	In panel	Excluded, absent	Pipeline matched	Control determ.	Genuine wedge	Reinstated (prior)
United States	1,603	0	1,603	946	687	360	946
Sweden	216	0	216	123	114	114	79
Subtotal	1,819	0	1,819	1,069	801	474	1,025

Note: Series, bases, and caveats as defined in Section 2.5. The two markets whose excluded multiclass firms have been reinstated under a named prior certified build (United States, *b1-us-append-v3.1*; Sweden, the cross-firm analog). *Pipeline matched* counts excluded firms matched to a derivation record (a bridged representative CIK node for the United States, a dual-class verdict row for Sweden); *control determination* the subset with an actual control determination; *genuine wedge* the subset adjudicated to carry a voting–cash-flow wedge. The 946 United States figure counts representative CIK nodes appended, bridged from 1,603 excluded issuers; for Sweden 216 is the raw exclusion count and 123 the pipeline-matched count, of which 79 reinstate cleanly. The full census across all markets, with panel definitions and per-market figures, is Table 22 in Appendix A.

The full census (Appendix A) sorts the remaining markets into four groups: the 33 markets whose excluded firms are matched to the dual-class verdict pipeline but not yet appended (Panel B, led by China, Canada, the United Kingdom, and Poland, with 440 verdict-matched firms of which 129 carry a genuine voting–cash-flow wedge but only 179 currently have a built voting-control vector), six markets whose excluded firms are recorded by the screen but not yet derived (Panel C, 78 firms led by Australia and Japan), markets carrying a dual-class pipeline with zero recorded exclusions (Panel D), and markets not yet scoped (Panel E). The census shows that the reinstatement covers the largest and most wedge-dense excluded populations and bounds the pending work by enumerating a defined set of firms with derived data awaiting a build or an append and a defined set of markets awaiting derivation.

The enlargement reverses only the co-authors' multiclass screen. Firms removed by their other sample screens remain out of both the baseline and the enlarged panels. Section 5.0.0.2 states the full screen inventory and its interpretive caveats.

8.3 The complete disposition of the excluded universe

The exclusion bias measured in Table 9 is the object of this section, and its interpretation depends on how much of the excluded universe the reinstatement has actually reached. That reach is stated completely in Table 12, which assigns every one of the 3,620 excluded issuers exactly one disposition and partitions the universe into six classes, so the measured bias is read against an explicit account of what remains unmeasured. The account is deterministic: each firm's status follows from the artifacts of the reinstatement build under a fixed precedence, with no matching judgment introduced at this stage.

The reinstatement has restored 1,104 firms through the certified route, comprising the 946 representative United States CIK nodes bridged from that market's 1,603 excluded issuers, the 79 Swedish firms of the register build, and the 59 firms of the certified market wave (whose round-two additions, three Polish, eight Canadian, and two Swedish firms, carry vectors and appends that passed an adversarial cross-model audit on 2026-07-22). A further 216 firms enter through two tagged, non-citable variants described below. The in-panel overlay carries the 306 firms that reappear in the panel through a per-security overlap, of which 245 already hold a mapped correction vector. The evidenced-infeasible class of 788 firms records a terminal outcome supported by evidence rather than an open task; it is dominated by the United States bridge shortfall of 637 issuers, which itself resolves into two evidenced stages, the 509 issuers with no usable 2020–2024 proxy statement and the 128 with no CIK bridge, and it includes one firm whose frozen wedge verdict failed current-vintage verification (Alimentation Couche-Tard, whose 2025 conversion collapsed the dual class). The remaining mass is prospective: 169 firms are in progress toward a build or an append, and 1,037 historical-vintage issuers, which fall outside the 2020–2024 estimation window, are shelved pending the vintage decision recorded in Section 12.

The certified append engine has now been executed across twenty Panel-B markets, the nineteen-market B1 wave together with the Sweden build, reinstating 138 Panel-B firms (81 for Sweden and 57 across the other markets) alongside the 946 bridged United States nodes. The Swedish build was enlarged beyond its certified 79 firms by a coverage-gap harvest of Telefonaktiebolaget LM Ericsson, whose controllers hold 24.5% (Investor AB) and 15.1% (Industrivärden) of the votes against 9.3% and 2.6% of the capital, and by a web-adjudicated identity recovery (Midway Holding); the certified 79 firms reproduce with identical control vectors inside the enlarged series. A third recovered identity, the merger-successor match of the pre-2004 Kinnevik entity to today's Kinnevik AB, is recorded in the spine but its vector remains held back by the over-sum

Table 12: Complete disposition of the excluded multiclass universe

	Firms (N)	Share
Panel A. Disposition class		
Reinstated (certified route)	1,104	30.5%
Reinstated (tagged variant, non-citable)	216	6.0%
In-panel overlay correction	306	8.5%
Evidenced infeasible	788	21.8%
In progress	169	4.7%
Historical-vintage, pending	1,037	28.6%
Total excluded universe	3,620	100.0%
Panel B. Status detail (within class)		
Reinstated		
United States, certified (966 nodes)	966	26.7%
Sweden, certified	79	2.2%
B1 market wave	59	1.6%
Reinstated (tagged variant)		
Control-chains overlay variant	169	4.7%
Proportional equal-vote variant	47	1.3%
In-panel overlay		
Correction vector mapped	245	6.8%
Correction vector pending	61	1.7%
Evidenced infeasible		
United States bridge shortfall	637	17.6%
No control-chain, no TR identity	110	3.0%
Control-chains vector infeasible	21	0.6%
Capital-only register	16	0.4%
Equal-vote, dispersed-only roster	3	0.1%
Verdict failed current-vintage verification	1	0.0%
In progress		
Harvestable, queued	97	2.7%
Matched, awaiting build/append	56	1.5%
Round-2 evidence, needs review	8	0.2%
Identity under review	8	0.2%
Historical-vintage, pending		
Historical-only vintage, shelved	1,037	28.6%

Market	Excl. (N)	Rein-stated	Var-iant	Over-lay	Infea-sible	In prog.	Hist. pend.
Panel C. Largest excluded populations by market							
United States	1,603	966	0	0	637	0	0
China	409	1	136	150	23	0	99
Canada	373	15	20	1	60	38	239
Sweden	216	81	0	0	0	20	115
United Kingdom	179	1	6	27	8	39	98
Poland	137	6	5	0	2	18	106

quarantine. The wave's Danish and Finnish appends rest on a web-grounded owner-identity adjudication that resolves the ownership registers' synthetic holder strings to the co-authors' Thomson–Reuters owner codes; the adjudication is anchored on two verified appends, the Danish holding behind Ambu and the Finnish pension holding behind Metsä Board and KONE, and once the verified owners replace the synthetic strings the Danish and Finnish reinstated levels rise above the synthetic-holder counterfactual the audit had flagged.

The two tagged variants extend coverage into markets and firm types the voting-derivation pipeline does not yet reach, and both are held strictly non-citable pending co-author adoption. The control-chains variant appends 169 firms across nineteen markets, carrying the ORBIS ultimate-owner divergence between control and cash-flow rights through pyramids and holding chains; that divergence is a full-strength instrument for the pyramid-control channel while serving only as a proxy for a per-class super-voting wedge, so the layer is a diagnostic rather than a substitute for a voting derivation. The proportional equal-vote variant appends 47 Chinese equal-vote firms under the votes-equal-capital identity, which is exact for genuinely equal-vote structures; a further 48 of the equal-vote firms are struck by the engine's in-panel guard because they already appear in the built panel, and 3 carry only a dispersed roster with no identified holder to append.

Read against this disposition, the exclusion bias of Table 9 is measured on the firms the reinstatement reaches, while the classes that remain open bound what the current measurement cannot yet see: chiefly the 1,037 historical-vintage issuers outside the estimation window and the 788 firms whose evidence records a terminal outcome, most with no per-holder voting leg to restore.

9 Voting–cash-flow separation among excluded US firms

The co-authors' panel excludes multiclass firms through a two-vintage bridge of 946 representative US CIK nodes (issuers sharing a CIK are collapsed to one node). Of these, 702 are in-scope multiple-class candidates whose control cannot be inferred from one-share-one-vote ownership. The remaining 244 fall outside the derivation's scope; most carry a single economic common class, and the census table details the remainder. For the scoped firms, we use SEC DEF 14A proxy statements to reconstruct each holder's voting-control share γ (the share of aggregate votes across all classes) and economic-ownership share β (the holder's cash-flow share). The construction of β weights each class separately, using zero for non-economic voting stubs, the as-exchanged figure for umbrella-partnership (Up-C) units, and the as-disclosed or as-converted amount for overlapping or convertible holdings; shares acquirable within 60 days are subtracted from both legs, and overlapping ownership rows are retained as the proxy discloses them. The voting–cash-flow wedge is $\gamma - \beta$, positive exactly when a holder's votes exceed the corresponding cash-flow rights.

Each scoped firm is classified from the certified reading of its actual voting terms into one of four verdicts: a *genuine wedge* when a super-voting class makes voting rights diverge from cash-flow rights, *equal-vote multiclass* when two or more economic classes carry equal votes, a *single-class artifact* when the second class is authorized-but-unissued, converted away, or otherwise not outstanding at the record date, and *documented unresolvable* when the proxy proves a class exists but never discloses the datum needed to settle its structure. Under the three-tier extraction of

Section 3.4, deterministic parsing handles jurisdiction-invariant mathematics. A bounded model-adjudication tier handles the residual judgment cases (voting-terms extraction, footnote linkage, control-group aggregation, and super-vote conventions); its decisions are cached, provenance-tagged, and cross-model audited before use.

Table 13: The US wedge universe: structural census and controller provenance

	N	%
Panel A. Structural census (excluded US multiclass paper universe)		
US multiclass paper-bridge CIKs (two vintages)	946	–
Unscoped: dilution-only, outside adjudication scope	244	–
Scoped and structurally adjudicated	702	100.0
Genuine voting–cash-flow wedge	360	51.3
Equal-vote multiclass (no wedge)	261	37.2
Single voting class (vendor multiclass artifact)	66	9.4
Documented unresolvable	15	2.1
Panel B. Resolved-controller extraction provenance (production controller dataset)		
Deterministic table parsing	1,143	76.4
Model-adjudicated (LLM)	278	18.6
Concert / group aggregation	76	5.1
Total resolved-controller rows	1,497	100.0
(memo) Flagged for human triage	51	3.4

Note: Series and audit conventions follow Section 2.5. Panel A reports the certified structural census of the US multiclass firms excluded from the co-authors' panel. The two-vintage bridge holds 946 CIKs (966 issuer-code rows), of which 702 are scoped for structural adjudication and 244 lie outside classification scope: 222 whose vendor multiclass flags resolve to a single economic common class, 7 with no extractable class structure, and 15 golden-benchmark firms reserved for validation and excluded from the derivation (among them Meta, Alphabet, Ford, and Berkshire Hathaway). Of the 702 scoped firms, 360 (51.3%) are genuine wedges; the remaining 342 comprise 261 equal-vote firms, 66 single-class artifacts, and 15 documented but unresolvable structures. Panel B reports the extraction provenance of a distinct companion artifact, the production resolved-controller dataset, at row grain (1,497 rows: 1,484 distinct nonblank accessions, 11 duplicate rows, and 2 with a blank accession): deterministic table parsing resolves 1,143 rows, a model-adjudication tier 278, and concert/group aggregation 76, with 51 flagged for human triage.[a]

[a]The production QA report (*qa_report.md*) records a marginally different method mix (1,144 deterministic / 277 LLM / 76 group) and 65 triage accessions; the one-row difference is the Masimo Corp filing (accession 0001104659-25-027887), classified deterministic in the QA report and model-adjudicated in the resolved-controller dataset.

Table 14 reports the largest wedges among the firms classified in the census. Across the dataset, a founder, family vehicle, or insider bloc often holds a large majority of the votes with a small share of the economic interest.

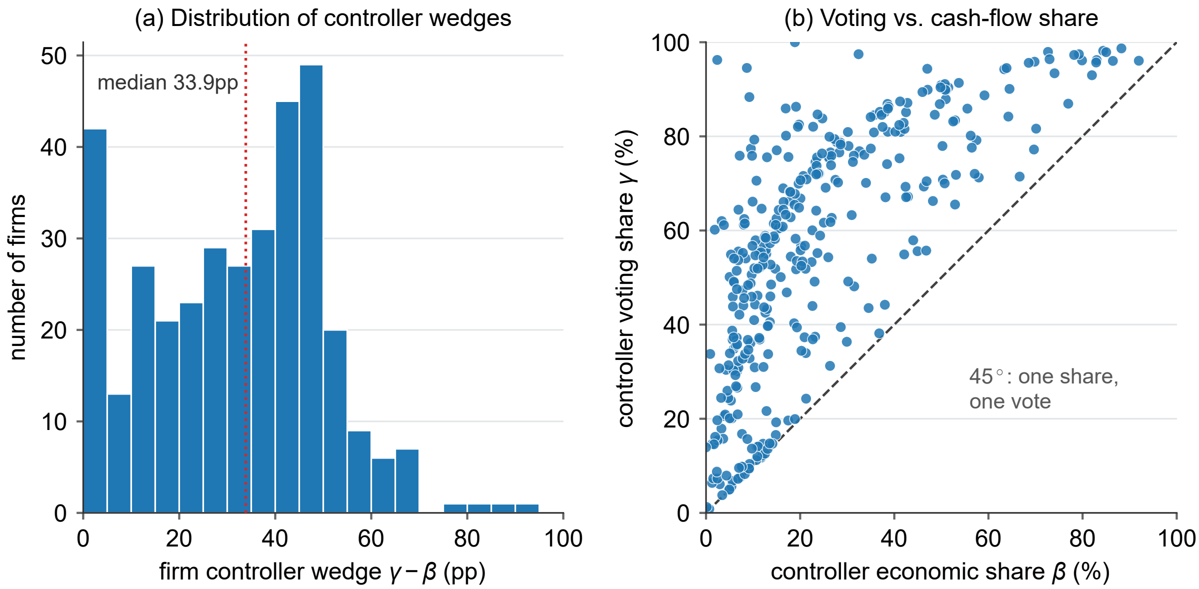
Table 14: Largest voting–cash-flow wedges among excluded US firms

Company	Controlling holder	Voting γ (%)	Economic β (%)	Wedge (pp)
MOLSON COORS BEVERAGE…	4280661 Canada Inc.	96.2	2.4	93.9
KELLY SERVICES INC	Terence E. Adderley Rev…	94.5	8.7	85.9
THE BOSTON BEER COMPA…	C. James Koch	100.0	18.9	81.1
Hims & Hers Health, I…	Andrew Dudum	88.3	9.2	79.1
DOMO, INC.	Joshua G. James	79.3	10.3	69.0
Expensify, Inc.	Expensify Voting Trust	85.9	16.9	69.0
ALPINE 4 HOLDINGS, IN…	All directors and execu…	75.9	7.1	68.7
Paramount Global	National Amusements, In…	77.4	9.5	67.9
UiPath, Inc.	Daniel Dines	86.3	19.1	67.2
Coupang, Inc.	Bom Kim	75.9	9.8	66.1
Oppenheimer Holdings…	A.G. Lowenthal	97.5	32.4	65.0
AppLovin Corporation	Shares subject to the V…	80.1	17.0	63.1
Coinbase Global, Inc.	All executive officers…	82.5	19.8	62.7
MARCHEX INC	Russell C. Horowitz	75.6	13.0	62.6
Clear Secure, Inc.	All directors and execu…	82.0	19.5	62.5
Duolingo, Inc.	All executive officers…	77.0	15.0	62.0
Hyperfine, Inc.	Jonathan M. Rothberg, P…	84.7	23.7	61.0
Coca-Cola Consolidate…	J. Frank Harrison, III,…	70.6	10.7	59.9

Note: The table reports the 18 excluded US firms with the largest certified voting–cash-flow wedge $\gamma - \beta$, drawn from the 360 genuine-wedge firms in the census (Table 13). The definitions of γ, β, and the wedge, together with the audit conventions, follow Sections 3.4 and 2.5. The controlling holder is the certified beneficial owner or voting bloc with the largest wedge in the firm's most recent DEF 14A; where a row names an aggregate ("all directors and executive officers as a group," "shares subject to the voting agreement"), the figure covers that as-disclosed bloc as a whole. Company and holder labels reproduce those in the certified derivation and are truncated to fit. Each row traces to a cited SEC accession recorded in the derivation dataset. Certified controller values are authoritative and can differ from a proxy's headline voting percentage, which often folds in 60-day options or uses a holder-specific denominator. Classic controlled firms such as Meta, Alphabet, Ford, and Berkshire Hathaway are golden-benchmark validation firms excluded from this derivation and do not appear here.

Table 14 reports the upper tail, and Figure 7 shows the full distribution. Of the 360 genuine-wedge firms, 357 have a computed maximum and 353 have a positive computed wedge. Among those 353, the median controller holds 33.9 percentage points more of the votes than of the cash flows, and the separation reaches 93.9 points at the maximum, and substantial separation also occurs outside the upper tail.

Figure 7: Distribution of voting–cash-flow wedges among genuine-wedge US firms



Note: The population comprises the 360 certified genuine-wedge US firms; γ, β, and the wedge are as in Section 3.4. A firm has a *computed maximum* when at least one holder in its certified legs carries both a γ and a β value (357 of 360), and a *positive maximum* when the largest holder-level wedge exceeds zero (353 of 357); the figure plots those 353 firms. Inclusion depends on holder coverage in the certified legs, regardless of whether the controller is itemized in the Item 403 table. Panel (a) shows the distribution of the controller wedge $\gamma - \beta$ in percentage points (median 33.9, maximum 93.9, Molson Coors). Panel (b) plots each firm's controller voting share γ against economic share β; the dashed 45° line is one share, one vote, and points sit above it by the wedge. Controllers routinely hold majority or near-majority votes (γ above 50%) on economic stakes of 10–40%, a separation that the control correction restores and a one-share-one-vote measure assumes away.

For these firms, a one-share-one-vote reading of the ownership register misstates the controller's power by tens of percentage points of the vote. The dataset provides a standalone census of the excluded US universe and characterises the voting structures of its 702 in-scope multiple-class candidates; the same 946-node bridge is added back to the panel in the reinstatement exhibit.

10 Attribution of the control correction by record grade

The control correction alters country-average profit weights $\bar{\kappa}$ by re-weighting recorded holdings and by adding newly discovered controllers. Re-weighting occurs when a firm's votes are reassigned to the controller identified by the traced ownership chain, even though every holding was already present in the Thomson Reuters data. Discovery adds controllers omitted from the Thomson Reuters roster that appear only after following an ownership chain to an ultimate owner the vendor did not link to the firm. Re-weighting uses holdings the paper already trusts, whereas each newly introduced owner identity requires separate verification.

The series displayed throughout this document is the headline series (the MAIN$\cup$SECONDARY control assignment), which admits only discovered controllers that survive verification. It re-weights holdings through the traced control chains, folds in the dual-class voting leg, and carries the verified and unverified discovered-controller tiers graded below. Across the 49 markets its 2024Q4 median movement is −0.83%, and in the United States it raises $\bar{\kappa}$ from 0.353 to 0.361 (a +2.28% move whose sign is positive). Earlier drafts instead displayed a permissive construction that retained every discovered-controller candidate record irrespective of grade; this draft demotes it to a labeled variant, the demoted permissive variant. Two independent findings provide the evidentiary basis for the demotion. The record-grade decomposition below shows that candidate records the Thomson Reuters roster cannot corroborate account for almost all of the demoted permissive variant's large movement, while the arithmetic audit that follows shows that the permissive construction double-counts even the verified controllers against their own Thomson Reuters rows.

Each candidate record asserts that a named owner controls a named company, and the correction's roughly 47,000 residual records are graded by evidentiary strength into 165 verified discovered controllers (internally MAIN), 1,352 unverified vendor-sourced controllers (internally SECONDARY), and 45,838 quarantined identities the roster could not corroborate (internally QUARANTINE), as defined in the record-grade conventions (2.5). The verified and unverified tiers the headline admits carry a combined discovered-controller denominator mass of about 113 against roughly 1,490 for the quarantined tier, so the mass the headline declines to admit is more than ten times the mass it admits.

10.1 Attribution by record grade

Table 15 admits a different set of discovered-controller records into the 2024Q4 correction in each row and reports the all-country median movement of $\bar{\kappa}$. Differences across rows identify the movement attributable to each record grade.

10.2 The permissive construction double-counts the discovered controllers

The record-grade decomposition explains why the demoted permissive variant moves so far, while an independent adversarial audit of the netting engine establishes a second, arithmetic reason to demote it. When a discovered controller is admitted, the intended correction is to *replace* the controller's existing Thomson Reuters rows with a single traced control leg without enlarging the cash-flow denominator. Across the 165 verified anchors on 155 issuers, the headline MAIN ∪ SECONDARY engine makes this replacement by removing the matched Thomson Reuters rows and installing the traced leg, touching a denominator mass of 79.96, a net change of +4.10 over the baseline those rows had carried. For the same 165 anchors, the permissive construction adds each discovered controller as a synthetic leg beside the rows it was meant to correct. It adds a synthetic mass of 86.24 while retaining 75.86 of matched Thomson Reuters denominator mass, touching a combined 162.09 where the headline engine touches 79.96, so the verified controllers are counted twice before any question of record quality arises.

The same audit confirmed that the headline engine's replacement has no unresolved overlaps. A full scan of all 1,381 unverified vendor-sourced records against the current Thomson Reuters roster returned zero name collisions and identified a single hand-correction overlap involving

Table 15: Attribution of the $\bar{\kappa}$ correction, by record grade (2024Q4, all 49 countries)

	Discovered records	Median 2024Q4 $\bar{\kappa}$ move (%)	Status
Demoted permissive variant (chains leg)	47,355	−38.24	demoted variant
No-discovery outer bound: TR-mapped controllers only	0	+0.71	outer bound
The strict series: verified discovered controllers	165	+0.71	verified tier
+ unverified vendor-sourced controllers	1,517	−0.76	headline CC leg; disclosed error

Note: Each row admits a different set of discovered-controller records into the 2024Q4 control-chain correction; the movement column is the all-49-country median percent change in country-average $\bar{\kappa}$ relative to the one-share-one-vote baseline, on the R15 netting basis. Rows add records cumulatively: *the no-discovery outer bound* admits only controllers carrying a hard Thomson Reuters mapping, *the strict series* adds the 165 verified anchors, and the headline control-chain leg further adds the 1,352 unverified vendor-sourced records; *the demoted permissive variant* instead retains all ~47,000 residual records ungraded. The two verified-only series move the median only +0.71% and the verified-plus-unverified series moves it to −0.76%, both inside the graded bracket [−0.7575%, +0.7057%], whereas the ~45,800 quarantined records the roster could not corroborate account for nearly all of the demoted variant's chains-leg −38.24% movement.

a Spanish family aggregate that included a spouse already present in the roster; the R15 layer moved the overlap to the quarantined tier. A companion arithmetic backstop applied across all 49 markets reverts to baseline any issuer-quarter whose Thomson Reuters and retained synthetic denominator masses would together exceed unity.

10.3 Measured quality of the unverified tier

A fresh local-language recheck measured the unverified tier's error at 26.0% (one-sided 95% upper bound 38.1%), with the demotions falling into a dozen error classes dominated by false-person attributions, superseded controllers, and custodial or state-institution rollups. A stress simulation that demotes random shares of the unverified vendor-sourced records up to 40% (beyond the measured upper bound) leaves every draw inside the graded bracket and moving toward the strict series as more records are removed, so no demotion rate the measured error could imply pushes the headline's control-chain leg outside the bracket. The verification census, the error taxonomy, and the stress simulation are reported in Appendix B, which is transitional pending the post-ship jurisdiction verification round.

Records the roster could not corroborate account for nearly all of the demoted permissive variant's chains-leg −38.24% median movement (the composed permissive display, which folds in the dual-class leg, moves −38.26%). Every graded series, whether strict or headline and whether stressed or unstressed, moves the median by less than one percent in either direction, inside a graded bracket that runs from verified records only (+0.71%) to verified plus unverified (−0.76%).

The displayed headline therefore admits only the graded discovered controllers and replaces their Thomson Reuters rows without retaining them; the demoted permissive variant is retained and explained here. This choice is made subject to co-author ratification, and the grading shows that the headline correction mainly re-weights holdings already trusted by the data, while genuine discovered controllers roughly offset in the aggregate.

11 Robustness of the control correction to modelling choices

The corrected common-ownership series rests on three modelling choices: how control is propagated through pyramid chains, how loyalty-voting regimes are treated, and how dual-class voting rights are attributed. The corrected country-average profit weight $\bar{\kappa}$ is insensitive to the control-propagation rule. Restoring the loyalty-voting firms that the headline quarantines moves $\bar{\kappa}$ by at most about 0.001, in Italy and the Netherlands alone. The dual-class attribution rule is material in only two markets, Brazil and Sweden, within the standalone dual-class satellites in which it is tested. All numbers are generated by *make_robustness_r15.py* from the R15 composed panel (*r15_composed_panel.parquet*), which carries each robustness leg as a column on the loyalty-quarantined MAIN ∪ SECONDARY headline basis.

11.1 Control-propagation rule: ultimate-owner trace vs. weakest-link

The headline series retains a controller identified through an ultimate-ownership trace, whereas the Aminadav–Papaioannou weakest-link rule breaks a chain at its weakest intermediate layer. The choice is immaterial at the country-average level in Table 16: the median absolute difference across the 49 markets is 0.0001 in $\bar{\kappa}$ and the largest anywhere is 0.0024 (Ireland).

Table 16: Control-rule robustness: ultimate-owner trace vs. weakest-link

	Ultimate-owner trace		Weakest-link			Δ% vs. baseline	
	2023	2024Q4	2023	2024Q4	Rule Δ (2024Q4)	Trace	Weak
Ireland	0.1124	0.1668	0.1101	0.1644	0.0024	−17.7	−18.9
United States	0.1686	0.2822	0.1675	0.2805	0.0017	−19.9	−20.4
South Africa	0.0775	0.0583	0.0795	0.0597	−0.0014	−67.5	−66.7
Morocco	0.0118	0.0121	0.0123	0.0129	−0.0008	−40.6	−36.6
Colombia	0.0104	0.0107	0.0111	0.0115	−0.0008	−31.6	−26.5
Egypt	0.0459	0.0483	0.0464	0.0491	−0.0008	−29.7	−28.6
China	0.0332	0.0299	0.0337	0.0305	−0.0006	−32.6	−31.2
Chile	0.0145	0.0139	0.0149	0.0142	−0.0004	−29.1	−27.2
Peru	0.0046	0.0072	0.0050	0.0076	−0.0004	−55.7	−53.4
United Arab Emirates	0.0238	0.0182	0.0243	0.0186	−0.0004	−36.9	−35.7
Austria	0.0092	0.0092	0.0095	0.0095	−0.0003	−51.1	−49.3
Finland	0.0244	0.0265	0.0246	0.0267	−0.0003	−40.2	−39.6
Netherlands	0.0331	0.0472	0.0332	0.0475	−0.0003	−32.4	−32.0
Japan	0.0236	0.1253	0.0236	0.1255	−0.0002	−41.3	−41.1
United Kingdom	0.0254	0.0282	0.0256	0.0284	−0.0002	−52.0	−51.7
Median, all 49	0.0111	0.0151	0.0112	0.0153	0.0001	−38.3	−36.6

Note: Country-average $\bar{\kappa}$ under two control-propagation rules; column, base-year, and roster-attrition conventions follow Section 2.5. The trace column is the headline rule, tracing ultimate ownership through pyramid chains (*combined_trace_DC_R1*); the weakest-link column applies the Aminadav–Papaioannou rule, under which a chain is only as strong as its weakest layer (*combined_weaklink_DC_R1*). Both legs are the demoted permissive variant, which the headline MAIN ∪ SECONDARY netting cannot supply for want of a weakest-link tier; because the rule difference is a property of chain tracing rather than of tier netting, the immateriality carries to the headline. "Rule Δ" is trace minus weakest-link at 2024Q4. Rows are the 15 countries of 49 with the largest absolute rule difference at 2024Q4; the final row is the all-49 median, whose "Rule Δ" cell is the median *absolute* difference, 0.0001 (maximum 0.0024, Ireland). Trace is presented as primary and weakest-link as robustness, pending a co-author decision.

11.2 Loyalty-voting treatment: headline quarantine vs. restored satellite

The headline quarantines the firms subject to loyalty-voting regimes, in which registered long-term holders receive additional votes, and a loyalty satellite restores them. Table 17 compares the two. The comparison is confined to the three markets with loyalty-treated firms in the panel; the remaining 46 markets are identical under both, so the all-49 median effect is zero. Restoring the loyalty firms lowers $\bar{\kappa}$ by at most 0.0011 (Italy) and 0.0009 (Netherlands) at 2024Q4, and by a negligible amount in France.

Table 17: Loyalty-voting robustness: headline quarantine vs. restored satellite

	Headline (quarantined)		Satellite (restored)			Δ% vs. baseline	
	2023	2024Q4	2023	2024Q4	Loyalty Δ (2024Q4)	Head.	Sat.
Italy	0.0062	0.0308	0.0059	0.0297	−0.0011	−3.0	−6.3
Netherlands	0.0452	0.0704	0.0447	0.0695	−0.0009	0.8	−0.5
France	0.0072	0.0129	0.0072	0.0129	0.0000	0.1	0.1
Median, all 49	0.0149	0.0260	0.0149	0.0260	0.0000	0.0	0.0

Note: Country-average $\bar{\kappa}$ under the headline, which quarantines loyalty-voting firms, and the loyalty satellite, which restores them (*combined_trace_DC_R1_loyaltyincl* on the headline-series basis); conventions follow Section 2.5. Under the adopted Option A the headline is loyalty-quarantined and the satellite is a supplementary display. "Loyalty Δ" is satellite minus headline at 2024Q4. Rows are the loyalty-treated markets whose satellite departs from the headline at the 2023 annual mean or 2024Q4, of 49; the final row is the all-49 median, whose cell is the median absolute effect. Only Italy and the Netherlands move at 2024Q4; France carries loyalty-treated firms earlier in the panel but none at 2024Q4, so its displayed effect is zero. Restoring the loyalty firms lowers $\bar{\kappa}$ by 0.0011 (Italy) and 0.0009 (Netherlands), because those firms carry the dual-class wedge the correction attributes to controlling holders; the effect is small, at most about 0.001, and overturns no levels or survival result.

11.3 Dual-class attribution rule: DC_R1 vs. DC_R2

Table 18 compares two rules for writing dual-class voting rights into the ownership register. The comparison uses standalone dual-class satellites with no ownership-chain correction, which isolates the attribution rule. Under the headline rule R1 a treated firm's control vector is replaced by the reconstructed voting-adjusted holders, whereas under R2 only the dual-class holders' rows are overwritten and all other owners persist. Within these satellites, the choice is material only in Brazil and Sweden at 2024Q4 and negligible everywhere else.

Table 18: Dual-class attribution rule: DC_R1 vs. DC_R2

	DC_R1 (headline)		DC_R2			Δ% vs. baseline	
	2023	2024Q4	2023	2024Q4	Rule Δ (2024Q4)	R1	R2
Brazil	0.0080	0.0710	0.0086	0.0786	−0.0076	−12.4	−3.0
Sweden	0.0421	0.0385	0.0433	0.0400	−0.0015	−5.2	−1.4
Austria	0.0090	0.0175	0.0097	0.0182	−0.0007	−7.0	−3.1
Germany	0.0120	0.0172	0.0123	0.0176	−0.0005	−4.3	−1.8
Korea	0.0016	0.0014	0.0019	0.0018	−0.0003	−25.7	−9.0
Israel	0.0140	0.0163	0.0141	0.0165	−0.0002	−1.3	−0.2
Finland	0.0385	0.0439	0.0387	0.0441	−0.0002	−0.7	−0.4
Italy	0.0061	0.0317	0.0061	0.0317	−0.0001	−0.2	−0.0
China	0.0475	0.0443	0.0475	0.0443	−0.0000	−0.0	−0.0
India	0.0012	0.0013	0.0012	0.0013	0.0000	0.0	0.0
Singapore	0.0016	0.0094	0.0016	0.0094	0.0000	0.0	0.0
Median, all 49					0.0000	0.0	0.0

Note: Corrected country-average $\bar{\kappa}$ under two rules for attributing dual-class voting rights; conventions follow Section 2.5. Both columns are the *standalone* dual-class satellites (*DC_R1* and *DC_R2*), with no ownership-chain correction, so only the attribution rule varies. Under R1 (headline) a treated firm's control vector is the reconstructed set of voting-adjusted holders; under R2 the baseline register is retained and only the dual-class rows are overwritten, so other owners persist and R2 stays nearer the baseline. "Rule Δ" is R1 minus R2 at 2024Q4. Rows are the *endpoint movers*, defined as markets whose R1 or R2 departs from the OSOV baseline at an endpoint, and are sorted by absolute rule difference; elsewhere R1 ≡ R2 ≡ baseline (final row: all-49 median, cell the median absolute difference). Of the twelve markets treated over 2020–2024Q4, eleven are displayed movers; Denmark, treated only in 2020 ($|R1 - R2| \leq 0.00008$), is not. The rule matters materially only for Brazil (0.0076) and Sweden (0.0015) at 2024Q4; Italy and the Netherlands collapse toward the baseline because their loyalty-linked dual-class firms are quarantined under Option A and surface in the loyalty satellite (Table 17).

The control-propagation comparison varies the full corrected series, and the earlier levels and survival results are insensitive to the propagation rule. The loyalty satellite moves $\bar{\kappa}$ by at most about 0.001, in Italy and the Netherlands, and leaves the headline conclusions intact. Within the standalone satellites, the dual-class attribution rule is material only in Brazil and Sweden. No combined series under rule R2 is computed, so that comparison does not test the combined levels and survival series.

12 Open decisions and next steps

The exhibits in Sections 5–11 present the control correction on a single chosen basis, the headline series (MAIN∪SECONDARY), alongside the verified-only strict series, the demoted permissive variant, and the specification robustness checks. This section records the editorial decisions taken in the present revision and the decisions and upstream tasks that remain open. The author made the decisions in the first list in his capacity as a co-author, subject to full co-author ratification. The figures throughout are drawn from the current build (the R15 netting basis) and remain subject to the cross-model audit before any number is cited.

The present revision made the following decisions, all of which remain reversible because labelled variants retain the alternatives. Ratification therefore amends a display choice without changing the computation.

1. **Displayed corrected series.** The displayed series is the headline series, MAIN∪SECONDARY, which combines verified discovered controllers with unverified vendor-sourced controllers. The vendor-sourced tier is admitted under benefit of the doubt with its measured error rate disclosed (26.0%, 95% upper bound 38.1%), while the held-out discovered-controller candidates (partly affirmatively refuted and partly unresolved; §3.6) are excluded from every displayed series. At 2024Q4 this estimator moves the all-country median $\bar{\kappa}$ by −0.83% and moves the United States from 0.353 to 0.361 (Section 6), replacing the former practice of displaying the permissive series.

2. **Strict verified-only bound.** The verified-only tier (MAIN, 165 records) is reported beside the headline as a strict robustness bound. It moves the all-country median $\bar{\kappa}$ by +0.71% at 2024Q4, so the sign of the median correction depends on whether the vendor-sourced tier is admitted; the two figures are therefore presented together.

3. **Demotion of the permissive variant.** The estimator displayed in earlier drafts (*combined_trace_DC_R1*), which retained all candidate records and moved the median by −38.26% at 2024Q4, is demoted to a labelled variant. The attribution exhibit (Section 10) retains and explains it as the transparency benchmark for how much of the earlier movement rested on records the hand-coded roster cannot corroborate. The overlap audit reported there supports the demotion.

4. **Loyalty-share treatment.** Loyalty-voting firms (33 issuers in France, Italy, and the Netherlands) are quarantined from the headline through the propagation gate and restored in a loyalty satellite. The robustness table T18 (Section 11) quantifies the difference; at the country median only Italy and the Netherlands move. This treatment implements Option A of the loyalty decision below. The upstream flag-propagation defect that had blocked this variant is fixed, and the combined build is available.

5. **Record de-duplication from the overlap audit.** Eight vendor-sourced records were demoted following the overlap audit: three unambiguous duplicates, four pension-plan records reclassified to the co-authors' institutional owner class (the Canada Pension Plan Investment Board, two Egyptian records, one Moroccan), and one Chinese record held pending identity resolution. The two New Zealand Superannuation Fund records are retained as a sovereign-wealth pair under the government-institution class, consistent with the depth-one government tier, a retention recorded in the present section and in the accompanying movement memo. The issuer-level arithmetic gate that reverts overfull issuers to baseline, generalised from the two Chinese cases that prompted it to an all-49-market class sweep (§3.6), applies here as the arithmetic backstop.

6. **Vintage anchoring of the corrected series.** Because the control overlay is of current (2020–2024) vintage, the corrected series is anchored to 2020Q1–2024Q4. A full 2005–2024 recompute gives the time-series exhibits a common headline basis and avoids splicing a permissive early window onto the recent one. The overfull-gate time profile (§3.6) quantifies the vintage incompatibility motivating this anchoring. As the panel runs back before 2020, the synthetic vendor-sourced vectors become arithmetically incompatible with historical holding sums in progressively more issuers.

The following decisions remain open and belong to the co-authors.

1. **Ratification of the series choice.** The configuration above (the headline series, the strict series, and the demoted permissive variant) is the author's recommendation. The co-authors must decide whether to adopt it formally or promote the strict series or the demoted permissive variant to the headline. Promoting the vendor-sourced tier further would require re-verification of its records against local-language primary filings; the verification corpus and method are retained so that re-verification can begin at any time.

2. **Control-propagation rule.** The corrected series traces ultimate ownership through pyramid chains and retains a controller identified through such a chain; the Aminadav–Papaioannou weakest-link rule is the alternative. Section 11 shows the choice to be immaterial at the country-average level, with a median absolute difference of 0.0001 in $\bar{\kappa}$ across the 49 markets and a maximum of 0.0024 (Ireland). The standing recommendation uses the ultimate-owner trace as primary and the weakest-link rule as a labelled robustness variant, since the weakest-link rule applies a second attenuation to the control side and deleting a pyramidal controller removes every common-ownership link it spans; formal adoption of this presentation is open.

3. **Loyalty-share regime.** The settled measurement mechanism is Option A: quarantine from the headline with a restoring satellite (decision item 4 above). The open sample-definition question is whether these loyalty firms ever enter the headline sample. The paper's original screen never removed them, so admitting them would extend the sample definition and should be a deliberate co-author choice. The recommended working default keeps them as an explicit opt-in satellite.

4. **Residual dual-class calls.** Two adjudication matters from the dual-class layer require co-author decisions. First, three wedge verdicts remain genuinely uncertain: one Indian issuer whose depository-receipt flag conflicts with a partly-paid-share reading, and two United Arab Emirates listings whose literal super-vote classes may be de-SPAC artifacts. Short evidence dossiers remain the pending deliverable and will inform the decision; the recommended default keeps all three outside the corrected set until a dossier positively confirms a genuine wedge. Second, the treatment of non-voting preferred classes (Korean preferred shares, Brazilian PN, German Vorzüge, Peruvian investment shares) follows the working convention of inclusion with statutory zero votes, with a voting-common-only variant computed alongside; ratification of that convention is open, and the recommendation is to retain it because the proportional-control assumption fails in these structures, while the voting-common-only variant already covers the narrower reading. Both matters can be implemented as one-line toggles

in the build without re-extraction.

5. **Formal regime classification.** The market ranking in Section 6 uses illustrative groups of four pyramid and four dispersed-ownership anchors. Its note records that the groups overlap in the middle of a continuous ordering. Whether the paper adopts a formal pyramid-versus-dispersed classification, and on what criterion, is a pending co-author decision; the levels result does not depend on any particular grouping in the figure.

6. **Attribution conventions to ratify.** The construction (Section 3) adopted a set of working conventions for cases in which the registers admit more than one defensible reading: whether a stake held through a personal holding vehicle is attributed to the vehicle or to the person behind it, and, in the United States derivation (Section 9), a catalogue of issuer-specific proxy conventions (aggregate-vote rules, as-converted and as-exchanged treatments, and the tolerance for divergence between beneficial-ownership disclosure and economic stakes). Each convention is documented with its adjudication record; the recommendation is to ratify the adopted defaults on the next call, revisiting individual conventions only where an exhibit turns on one.

A further set of decisions remains for the internal cross-model audit to resolve before any affected figure is cited: the **Barclays Wealth MAIN anchor** (issuer code 1355025), a scope violation that would alter a verified-tier record and so was not acted on in the present build; the **pre-2020 survival display**, a choice between the mirrored, vintage-flagged corrected line and the clean ownership-chain line for the window before 2020 in which the channel legs are not separately isolable; three **engine-hardening** corrections from the overlap audit (an owner-role guard against reading an asset-manager or institutional row as a controller, porting the *q1* over-sum gate into the *s4_recompute* stage, and correcting the legacy label that still calls the demoted permissive branch "corrected"); and the **Chinese overfull attribution**, where two issuers caught by the arithmetic gate require manual adjudication of whether the state-tracing overlay and the co-authors' state-owned-enterprise consolidation double-count the same control.

The remaining work concerns upstream data, code, and post-shipping verification.

1. **Jurisdiction verification of the parked mass.** The quarantined records are overwhelmingly unresolved; few have been affirmatively refuted. The headline is therefore a known-direction lower bound with respect to discovery because genuine controllers may remain parked for want of verification. A consequence-ranked verification round, taking Singapore, Hong Kong, India, Indonesia, and the Philippines first, is planned after shipping to promote the records that verification confirms.

2. **United States dual-class merge.** The US DEF 14A voting vectors are delivered as a labelled panel-enlarged variant (*combined_trace_DC_R1_plusUSreinst*, Section 8), which composes the headline correction with the reinstated excluded United States firms and is recomputed jointly over the enlarged firm set. The integration overlay itself still carries no United States dual-class rows, so the United States enters the census and levels exhibits through ownership-chain tracing only (Sections 5 and 6). The co-authors' pending sample-definition decision concerns

whether to adopt the enlarged panel as the default and, if adopted, merge the vectors into the overlay and rerun the affected exhibits.

3. **Reinstatement rollout beyond the two pilots.** The multiclass exclusion is panel-wide (3,620 issuers across 41 markets, Table 22), and the complete disposition of every excluded firm is now recorded in Table 12. The certified append has been executed across twenty Panel-B markets, reinstating 138 Panel-B firms alongside the 946 United States nodes, and two tagged non-citable variants add a further 216 firms. Completing the rollout requires building voting-control vectors for the remaining verdict-matched Panel B firms, harvesting Poland's cashflow-leg residual (where 14 of its 20 genuine-wedge firms still lack a per-holder cashflow leg), recovering the two South African super-voting firms through a share-ratio harvest, and scoping the markets where exclusion or pipeline coverage is not yet determined (Panels C and E). The 35 built-but-unappended firms that carry a two-leg vector under a non-genuine verdict (Volvo, Investor AB, and similar incumbents) are a staged candidate for the next append wave. This rollout is the stated limitation of the reinstatement analysis and its priority follow-on work.

4. **Round-2 harvest residuals.** The 2026-07-22 harvest wave leaves four scoped residuals: the Kinnevik AB vector is held back by the over-sum quarantine and needs a holder netting pass; five web-matched Swedish firms (IFS, Iptor, Precio Fishbone, Ortivus, Slottsviken) carry adjudicated identities but no wedge adjudication or built vector; the Peru build ships Alicorp under the new Inversión-inclusive denominator while 23 further genuine-wedge firms have verified class totals but no usable per-holder controller holdings to pair with them; and Fairfax Financial is parked on the interaction between its 41.8% voting cap and the uncapped 50 : 1 share terms, which needs an explicit modeling decision before its vector can ship.

5. **Reinstatement vintage target.** Whether reinstatement targets the 2020–2024 estimation window only or the full 2005–2024 history determines whether the 1,038 shelved historical-vintage issuers ever enter a κ panel. The working default adopts the complete-disposition framing tentatively, dispositioning every excluded firm within the current window, with a historical-vintage rebuild deferred to a co-author decision.

6. **Japan onboarding.** A proper Japanese voting derivation is blocked on a free EDINET API key from the Financial Services Agency. Japan is the one Panel C market whose excluded firms (13, Table 22) warrant a full derivation rather than a one-share-one-vote confirmation once the key is obtained; it is recorded here so the blocker stays in view.

7. **Exclusion-screen country scope.** The disposition covers the 41 markets present in the co-authors' delivered exclusion record. Russia and Romania are absent from that record and are a probable screen gap; confirming the screen's country scope and obtaining the raw multi-security Thomson–Reuters delivery would let the multiclass screen run for those markets.

8. **Regression-analog exhibits.** The cross-sectional counterpart to the survival analysis would re-estimate the co-authors' regression tables with corrected $\bar{\kappa}$ as a drop-in dependent variable,

reporting coefficients side by side. This awaits the co-authors' specifications and estimation data and the series-and-rule decisions above; the control-propagation alternative of Section 11 is the natural robustness variant.

9. **Voting-power robustness bracket.** Whether the co-authors' existing voting-power robustness checks span the register-based correction is an open computation, planned as a companion exhibit but not yet built.

10. **Uncertain-verdict dossiers.** The web-evidence dossiers for the three uncertain dual-class verdicts (open co-author item 4) are a pending deliverable feeding that decision.

11. **Further register reconstructions.** The register-based reinstatement analysis of Section 8 covers the United States and Sweden; extending it to the remaining markets with usable share registers, and to the equal-vote and artifact classes of the excluded universe, is scheduled after the decisions above.

The co-authors should first ratify the series choice and then settle the control-propagation rule, the loyalty-regime membership, the residual dual-class calls, and the working conventions. The internal audit resolves the Barclays anchor, the pre-2020 survival display, and the engine-hardening items before any figure is cited. The loyalty satellite, the US merge, and the regression analogs then proceed under the resulting configuration. A comprehensive cross-model verification of the present draft, extending to the complete-disposition accounting of Section 8 and the two tagged reinstatement variants, remains owed before any number in it is cited.

A Reference tables

The tables in this appendix provide reference material for selective consultation. The totals and argument-bearing numbers appear in the main text; each table below contains the full per-market detail behind a main-text excerpt or cross-market summary, and every note begins with the series, bases, and caveats defined in Section 2.5.

Table 19: Coverage and treatment of the control correction, by market

	Firms covered	Treated firms (by mechanism)			Total treated	Treated firms (%)	Treated mkt. cap (%)	Mean $\lvert\gamma-\beta\rvert$
		Chains	Dual-cl.	Both				
China	11979	3505	78	0	3583	29.9	52.5	0.158
United States	7195	2064	0	0	2064	28.7	17.4	0.085
India	6149	2464	3	2	2469	40.2	71.1	0.125
Japan	4024	1222	0	0	1222	30.4	25.1	0.122
Korea	2745	1141	36	63	1240	45.2	86.9	0.147
Canada	2475	426	1	1	428	17.3	12.9	0.124
Taiwan	2122	772	2	0	774	36.5	13.4	0.093
Australia	1763	731	0	0	731	41.5	16.7	0.080

	Firms covered	Treated firms (by mechanism)			Total treated	Treated firms (%)	Treated mkt. cap (%)	Mean $\lvert\gamma - \beta\rvert$
		Chains	Dual-cl.	Both				
Vietnam	1574	775	1	0	776	49.3	77.7	0.151
United Kingdom	1506	886	0	1	887	58.9	39.8	0.098
Malaysia	1074	644	0	0	644	60.0	63.7	0.158
Indonesia	953	522	0	0	522	54.8	59.8	0.222
Thailand	873	373	0	0	373	42.7	43.3	0.181
Sweden	836	217	83	54	354	42.3	–	0.128
Russia	746	331	0	0	331	44.4	71.1	0.204
Poland	715	373	0	0	373	52.2	69.4	0.156
France	628	351	13	2	366	58.3	–	0.202
Germany	609	315	5	10	330	54.2	70.0	0.208
Singapore	562	335	12	0	347	61.7	56.0	0.182
Israel	457	171	1	2	174	38.1	40.0	0.168
Brazil	449	192	75	49	316	70.4	72.3	0.234
Italy	381	270	4	14	288	75.6	70.5	0.206
Romania	338	163	0	0	163	48.2	30.7	0.194
Saudi Arabia	306	86	0	0	86	28.1	9.8	0.112
Philippines	276	61	0	0	61	22.1	29.9	0.212
Spain	272	191	0	0	191	70.2	63.8	0.193
Norway	268	172	0	1	173	64.6	35.2	0.142
Switzerland	239	85	0	0	85	35.6	35.0	0.132
South Africa	228	130	0	0	130	57.0	64.0	0.100
Hong Kong	226	111	0	0	111	49.1	61.0	0.168
Egypt	224	84	0	0	84	37.5	31.2	0.176
Jordan	193	81	0	0	81	42.0	65.6	0.134
Chile	177	119	0	0	119	67.2	83.7	0.330
Denmark	169	96	10	3	109	64.5	50.5	0.139
Finland	167	91	6	4	101	60.5	93.8	0.105
Peru	165	79	1	1	81	49.1	72.7	0.405
Netherlands	144	69	3	2	74	51.4	72.5	0.171
U.A.E.	143	69	1	0	70	49.0	55.5	0.246
Kuwait	134	35	0	0	35	26.1	11.2	0.222
Greece	132	48	0	0	48	36.4	21.8	0.132
Belgium	128	74	0	0	74	57.8	52.2	0.186
Oman	128	70	0	0	70	54.7	53.3	0.173
Mexico	127	60	0	1	61	48.0	69.8	0.230
New Zealand	124	58	0	0	58	46.8	31.9	0.077
Ireland	83	42	0	0	42	50.6	14.1	0.068
Croatia	82	26	0	0	26	31.7	25.1	0.127
Morocco	74	36	0	0	36	48.6	62.2	0.278
Austria	66	38	3	2	43	65.2	74.5	0.208
Colombia	60	38	0	0	38	63.3	69.1	0.361
All countries	54488	20292	338	212	20842	38.3	50.4	0.144

Note: Series, bases, and caveats are as defined in §2.5. Markets are ordered by the number of firms covered. *Firms covered* counts distinct listed firms for which the correction computes a corrected control vector; the census uses this cross-sectional *gamma_overlay* universe as its covered definition. It is not nested with the overlay panel's per-quarter universe, which covers more firms than the cross-section in some markets and fewer in others. *Treated firms* are those whose control vector the correction alters, identified from an owner-level wedge $\gamma - \beta$ between corrected control (γ) and cash-flow rights (β) exceeding a 10^{-6} tolerance: a firm is chains-treated when ownership-chain tracing moves control (traced control differs from the cash-flow chain), dual-class-treated when it carries a genuine voting–cash-flow wedge, and *both* when both mechanisms apply; dual-class firms are matched to their ownership-chain identity through the recorded partner identifier. *Treated firms (%)* is treated over covered. *Treated mkt. cap (%)* is the treated firms' share of covered-firm market capitalisation, computed within each market over its modal reporting currency; the All-countries entry is the unweighted mean of the market-level shares over the 47 joinable markets. A mechanical capitalisation-weighted global share would evaluate to 70.2% but would sum local-currency capitalisations without exchange-rate conversion. Two markets whose firm identifiers do not join to the market-cap source (France, Sweden) are shown as "–". *Mean* $|\gamma - \beta|$ is the largest owner-level control-share reallocation within a treated firm, averaged over treated firms, in control-share units. In this integration vintage the dual-class layer covers 22 markets, and the United States enters only through ownership-chain tracing. Its DEF 14A voting vectors are reported separately as a labeled panel-enlarged variant in the reinstatement exhibit of Section 8 and remain outside the overlay. The reinstatement effect of the two-vintage CIK bridge is quantified there and omitted from the integration-layer reinstatement table. At least 566 of the excluded U.S. multiclass issuers nonetheless enter the census's chains coverage through their ownership-chain identities. Panel cross-checks use the demoted permissive variant at reference quarter 2023Q4; the applied-under-headline counts on the R15 netting basis are recorded in *r15_applied_numbers.json*, and the full per-market figures, sources, and discrepancies in *census_numbers.json*.

Table 20: Overlap between the correction data and the co-authors' TR panel, by market

	TR-panel firms	Covered firms	Matched to TR	Match (%)	Treated firms	Overlay applied	Applied/ TR (%)	Treated cap (%)
China	5610	11979	4654	38.9	3583	3637	64.8	66.7
India	3947	6149	4348	70.7	2469	3580	90.7	81.1
United States	3757	7195	5454	75.8	2064	2935	78.1	95.6
Japan	3681	4024	3744	93.0	1222	3268	88.8	84.9
Korea	2440	2745	2283	83.2	1240	2137	87.6	95.2
Canada	1779	2475	2257	91.2	428	1177	66.2	69.0
Taiwan	1723	2122	1621	76.4	774	1545	89.7	80.7
Australia	1545	1763	1591	90.2	731	1212	78.4	75.4
Hong Kong	1219	226	193	85.4	111	140	11.5	22.9
United Kingdom	1205	1506	1198	79.5	887	900	74.7	92.4
Malaysia	918	1074	944	87.9	644	719	78.3	66.4
Vietnam	873	1574	125	7.9	776	92	10.5	15.6
Thailand	842	873	573	65.6	373	531	63.1	69.4
Indonesia	603	953	857	89.9	522	575	95.4	81.4
Poland	564	715	601	84.1	373	474	84.0	68.0
Singapore	531	562	479	85.2	347	396	74.6	54.0
Sweden	524	836	683	81.7	354	387	73.9	63.0
Israel	512	457	411	89.9	174	304	59.4	68.5
France	509	628	517	82.3	366	411	80.7	80.2
Germany	431	609	471	77.3	330	347	80.5	90.0
Italy	380	381	304	79.8	288	269	70.8	83.2
Romania	301	338	238	70.4	163	220	73.1	39.3

	TR-panel firms	Covered firms	Matched to TR	Match (%)	Treated firms	Overlay applied	Applied/ TR (%)	Treated cap (%)
Brazil	261	449	304	67.7	316	207	79.3	87.5
Saudi Arabia	247	306	6	2.0	86	5	2.0	0.3
Norway	230	268	226	84.3	173	184	80.0	93.6
Spain	224	272	203	74.6	191	169	75.4	81.2
Philippines	217	276	248	89.9	61	185	85.3	92.3
Egypt	205	224	154	68.8	84	137	66.8	75.6
Switzerland	197	239	210	87.9	85	164	83.2	91.5
South Africa	181	228	193	84.6	130	140	77.3	75.7
Jordan	178	193	38	19.7	81	31	17.4	41.3
Chile	147	177	147	83.1	119	101	68.7	76.1
Greece	146	132	89	67.4	48	80	54.8	62.5
Finland	145	167	156	93.4	101	118	81.4	96.6
U.A.E.	144	143	110	76.9	70	101	70.1	84.1
Kuwait	137	134	94	70.1	35	81	59.1	42.7
Russia	128	746	2	0.3	331	1	0.8	0.5
Netherlands	124	144	123	85.4	74	89	71.8	88.5
New Zealand	112	124	115	92.7	58	85	75.9	79.5
Denmark	101	169	133	78.7	109	76	75.2	76.7
Oman	101	128	94	73.4	70	77	76.2	87.4
Belgium	96	128	105	82.0	74	84	87.5	77.2
Croatia	75	82	51	62.2	26	36	48.0	44.0
Ireland	65	83	74	89.2	42	59	90.8	99.9
Morocco	64	74	52	70.3	36	46	71.9	84.1
Mexico	62	127	106	83.5	61	48	77.4	85.0
Austria	55	66	57	86.4	43	49	89.1	95.0
Colombia	54	60	51	85.0	38	40	74.1	73.6
Peru	49	165	106	64.2	81	41	83.7	72.5
All markets	37639	54488	36793	67.5	20842	27690	73.6	71.6

Note: Series, bases, and caveats as defined in Section 2.5. Market-level overlap between the correction datasets and the co-authors' TR holdings panel at the reference quarter 2023Q4. *TR-panel firms* is the overlay-panel firm count; *Covered firms* is the cross-sectional correction universe and *Treated firms* the covered firms whose control vector the correction changes (total 20,842), both identical to exhibit T1. *Matched to TR* counts covered firms whose crosswalk row matches a TR issuer at the ISIN, exact-name, or fuzzy-name tier, *Match (%)* scaling by covered firms. *Overlay applied* counts TR firms receiving the demoted permissive variant's corrected vector at 2023Q4 (total 27,690; the headline series applies to 27,174), *Applied/TR (%)* scaling by TR-panel firms. *Treated cap (%)* is the overlay-applied firms' share of TR-panel market capitalisation, the All-markets row giving the unweighted cross-market mean for this column and column sums otherwise; this share uses the TR-panel denominator and is not comparable to the census's covered-universe treated share (50.4%).

Table 21: Channel-isolated decomposition of the control correction, by market (2024Q4)

	Baseline $\bar{\kappa}$	Change in $\bar{\kappa}$ by channel				Combined (% of base)
		Chains	Dual-cl.	Inter.	Combined	
New Zealand	0.1116	−0.0135	0.0000	0.0000	−0.0135	−12.1
Brazil	0.0811	−0.0049	−0.0101	0.0046	−0.0103	−12.7
Mexico	0.0399	−0.0066	0.0000	0.0000	−0.0066	−16.5
Oman	0.1173	−0.0065	0.0000	0.0000	−0.0065	−5.6
Taiwan	0.0368	−0.0057	0.0000	0.0000	−0.0057	−15.4
Sweden	0.0406	−0.0034	−0.0021	0.0005	−0.0051	−12.4
Malaysia	0.0264	−0.0037	0.0000	0.0000	−0.0037	−14.1
Finland	0.0442	−0.0031	−0.0003	0.0000	−0.0034	−7.6
Canada	0.0307	−0.0026	0.0000	0.0000	−0.0026	−8.4
Kuwait	0.0313	−0.0025	0.0000	0.0000	−0.0025	−8.1
Morocco	0.0204	−0.0021	0.0000	0.0000	−0.0021	−10.3
Chile	0.0196	−0.0018	0.0000	0.0000	−0.0018	−9.4
Japan	0.2133	−0.0018	0.0000	0.0000	−0.0018	−0.8
Germany	0.0179	−0.0012	−0.0008	0.0003	−0.0017	−9.3
South Africa	0.1796	−0.0014	0.0000	0.0000	−0.0014	−0.8
Italy	0.0317	−0.0009	−0.0001	0.0000	−0.0009	−3.0
Norway	0.0212	−0.0008	0.0000	0.0000	−0.0008	−3.8
Romania	0.0199	−0.0004	0.0000	0.0000	−0.0004	−2.1
Israel	0.0165	−0.0002	−0.0002	0.0000	−0.0004	−2.4
Korea	0.0019	0.0001	−0.0005	0.0001	−0.0004	−18.1
Peru	0.0163	−0.0003	0.0000	0.0000	−0.0003	−1.9
Belgium	0.0234	−0.0003	0.0000	0.0000	−0.0003	−1.3
Spain	0.0065	−0.0003	0.0000	0.0000	−0.0003	−4.3
Croatia	0.0407	−0.0003	0.0000	0.0000	−0.0003	−0.7
Thailand	0.0034	−0.0002	0.0000	0.0000	−0.0002	−6.4
Colombia	0.0156	−0.0002	0.0000	0.0000	−0.0002	−1.0
Greece	0.0038	−0.0001	0.0000	0.0000	−0.0001	−1.9
Hong Kong	0.0165	−0.0001	0.0000	0.0000	−0.0001	−0.3
Singapore	0.0094	0.0000	0.0000	0.0000	0.0000	−0.5
Poland	0.0028	0.0000	0.0000	0.0000	0.0000	−0.7
Jordan	0.0260	0.0000	0.0000	0.0000	0.0000	−0.0
United Arab Emirates	0.0289	0.0000	0.0000	0.0000	0.0000	−0.0
Vietnam	0.0030	0.0000	0.0000	0.0000	0.0000	−0.1
Russia	0.0089	0.0000	0.0000	0.0000	0.0000	0.0
Saudi Arabia	0.0372	0.0000	0.0000	0.0000	0.0000	0.0
Indonesia	0.0004	0.0000	0.0000	0.0000	0.0000	2.2
India	0.0013	0.0000	0.0000	0.0000	0.0000	0.8
France	0.0129	0.0000	0.0000	0.0000	0.0000	0.1
Switzerland	0.0957	0.0001	0.0000	0.0000	0.0001	0.1
Philippines	0.0027	0.0001	0.0000	0.0000	0.0001	2.7
Denmark	0.0258	0.0002	0.0000	0.0000	0.0002	0.7
Egypt	0.0688	0.0002	0.0000	0.0000	0.0002	0.3
China	0.0443	0.0005	0.0000	0.0000	0.0005	1.2

	Baseline $\bar{\kappa}$	Change in $\bar{\kappa}$ by channel				Combined (% of base)
		Chains	Dual-cl.	Inter.	Combined	
Netherlands	0.0698	0.0006	0.0000	0.0000	0.0006	0.8
Australia	0.0345	0.0014	0.0000	0.0000	0.0014	4.0
United Kingdom	0.0588	0.0022	0.0000	0.0000	0.0022	3.8
Austria	0.0188	0.0045	−0.0013	0.0000	0.0031	16.7
United States	0.3525	0.0080	0.0000	0.0000	0.0080	2.3
Ireland	0.2026	0.0131	0.0000	0.0000	0.0131	6.5
All 49 (median)	0.0260	−0.0002	0.0000	0.0000	−0.0002	−0.8
All 49 (mean)	0.0476	−0.0007	−0.0003	0.0001	−0.0009	−3.1

Note: Series, bases, and caveats are as defined in §2.5. For each of the 49 markets in the κ panel at 2024Q4, the table reports the baseline one-share-one-vote average of the firm-level common-ownership measure and the change attributable to each correction channel taken separately: *chains* is ownership-chain tracing alone under the MAIN∪SECONDARY netting (*net_main_secondary*), *dual-class* is dual-class voting rights alone (*DC_R1*), and *interaction* is the combined change minus the two single-channel changes, so the combined column equals chains plus dual-class plus interaction by construction and the final column expresses it as a percentage of baseline. Because each firm is routed to a single mechanism per variant, the applied treated sets are disjoint and the interaction measures only the cross-channel κ links between a chains-treated and a dual-class-treated firm. Markets are ordered by the size of the combined correction; the two summary rows report the cross-market median and mean. The United States enters through ownership-chain tracing alone in this vintage, so its dual-class and interaction entries are an identified zero; full per-market provenance is recorded in *decomp_numbers.json*.

Table 22: Census of excluded multiclass firms and their reinstatement status, by market

Market	Excluded (N)	In panel	Excluded, absent	Pipeline matched	Control determ.	Genuine wedge	Reinstated (prior)
Panel A. Reinstated (prior certified builds)							
United States	1,603	0	1,603	946	687	360	946
Sweden	216	0	216	123	114	114	79
Subtotal	1,819	0	1,819	1,069	801	474	1,025
Panel B. Excluded multiclass pending reinstatement (matched to the dual-class verdict pipeline)							
China	409	150	259	100	99	1	0
Canada	373	1	372	46	43	19	0
United Kingdom	179	27	152	60	20	2	0
Poland	137	0	137	60	20	20	0
Indonesia	89	31	58	7	7	0	0
Mexico	71	6	65	10	0	0	0
Switzerland	53	14	39	11	5	5	0
Peru	47	0	47	29	25	21	0
Finland	44	18	26	20	13	13	0
Denmark	41	0	41	26	15	15	0
Netherlands	29	6	23	6	5	5	0
France	28	7	21	4	4	4	0

Italy	26	13	13	13	13	7	0
Philippines	26	0	26	3	0	0	0
South Africa	25	0	25	10	4	4	0
Belgium	21	0	21	0	0	0	0
India	21	16	5	12	12	5	0
Hong Kong	17	0	17	4	0	0	0
Israel	15	4	11	3	3	3	0
Chile	12	8	4	1	1	0	0
Norway	12	0	12	5	1	1	0
Germany	11	2	9	0	0	0	0
Spain	8	0	8	2	2	1	0
Singapore	8	0	8	3	3	2	0
Austria	4	0	4	1	0	0	0
Malaysia	4	0	4	1	1	0	0
Thailand	4	0	4	0	0	0	0
Brazil	3	1	2	1	1	1	0
Greece	2	0	2	1	0	0	0
United Arab Emirates	1	0	1	0	0	0	0
Colombia	1	1	0	0	0	0	0
Morocco	1	0	1	0	0	0	0
Oman	1	0	1	1	1	0	0
Subtotal	1,723	305	1,418	440	298	129	0
Panel C. Excluded multiclass identified; voting-data derivation not yet built							
Australia	45	0	45	0	0	0	0
Japan	13	0	13	0	0	0	0
Ireland	9	0	9	0	0	0	0
Croatia	6	0	6	0	0	0	0
New Zealand	4	0	4	0	0	0	0
Egypt	1	1	0	0	0	0	0
Subtotal	78	1	77	0	0	0	0
Panel D. Multiclass firms present in the pipeline; zero construction exclusions recorded							
South Korea	0	0	0	–	–	–	–
Taiwan	0	0	0	–	–	–	–
Vietnam	0	0	0	–	–	–	–
Panel E. No excluded multiclass firms recorded; market not yet scoped							
Jordan	–	–	–	–	–	–	–
Kuwait	–	–	–	–	–	–	–
Romania	–	–	–	–	–	–	–
Russia	–	–	–	–	–	–	–
Saudi Arabia	–	–	–	–	–	–	–
Total excluded universe	3,620	306	3,314	1,509	1,099	603	1,025

Note: Series, bases, and caveats as defined in Section 2.5. The excluded universe is the set of multiclass issuers the co-authors' sample construction removes from the estimation panel, taken from their construction-time exclusion file (3,620 distinct issuers across 41 markets, keyed on the Thomson-Reuters issuer code). *In panel* counts the excluded issuers that nonetheless appear in the built 2020–2024 panel through per-security overlap; *excluded, absent* is the complementary reinstatement universe. *Pipeline matched* counts the excluded firms matched to a derivation record. For the non-US markets, a match is a dual-class verdict row of any value; for the United States, it is a bridged representative CIK node. The 946 United States figure uses representative nodes as its counting unit because 1,392 of the 1,603 excluded US issuers bridge to a CIK, and the usable-proxy set links 966 issuer codes to 946 CIKs after collapsing the 20 CIKs that carry two codes each. *Control determination* counts the subset for which an actual control determination exists (a non-*no data* verdict for the non-US markets, and *status* = derived for the United States, 687 of the 946 nodes). *Genuine wedge* is the subset adjudicated to carry a voting–cash-flow wedge, a time-invariant verdict class distinct from whether an operational corrected vector was built or is active in any given firm-quarter (at 2024Q4, 61 of the 360 United States genuine-wedge nodes enter as dilution-only). *Reinstated (prior)* counts firms already appended under a named prior certified build (United States, *b1-us-append-v3.1*; Sweden, the cross-firm analog); the United States counts come from the certified skill-derived DEF 14A adjudication at representative-CIK granularity, whereas the non-US counts come from the firm-level dual-class *verdicts.csv*. Panel A holds the two reinstated markets. Panel B holds markets matched to the dual-class verdict pipeline but not yet appended, where verdict coverage runs ahead of operational readiness: of the 440 verdict-matched firms only 163 (and 79 of the 129 genuine-wedge firms) currently have a built voting-control vector, and nine Panel B markets carry verdict coverage but no built vectors, so extending the correction to them requires a build step before an append. Panel C holds markets whose excluded firms the co-authors' screen records but for which no dual-class derivation has been built, so their pipeline, determination, wedge, and reinstated counts are construction-imposed zeros and do not reflect an empirical finding. Panel D holds markets carrying a dual-class pipeline for which the co-authors' file records no construction exclusion; entity-level in-panel retention is verified for South Korea (97 of 102 verdict firms match a current-panel issuer code) but rests on pipeline presence with zero recorded exclusions for Taiwan and Vietnam, and the reinstatement columns do not apply. Panel E holds markets absent from the co-authors' exclusion screen and not yet covered by any dual-class pipeline; their counts are shown as not-determined (–). A zero is inappropriate because a genuine absence of multiclass firms cannot yet be distinguished from an unscoped market. For Sweden the co-authors' file records 216 excluded issuers, of which 123 match our dual-class verdict universe (the set on which reinstatement operated), and 79 reinstate cleanly; both the raw 216 and the pipeline-matched 123 are shown. Subtotals and the grand total cross-foot exactly (3,620 = 306 in panel + 3,314 absent; 1,025 reinstated = 946 United States + 79 Sweden). Full per-market figures, sources, and method notes are recorded in *table_B2census_numbers.json*.

Table 23: Panel-enlargement channel: reinstated multiclass issuers (2024Q4)

	Reference $\bar{\kappa}$	Enlarged $\bar{\kappa}$	Δ total	Incumbent movement		
				Total	Dilution	Linkage
United States	0.3606	0.2301	−0.1304	−0.0678	−0.0846	0.0168
Sweden (OSOV)	0.0406	0.0309	−0.0097	−0.0050	−0.0063	0.0013

Note: The United States rows are drawn from the R15 panel-enlarged build (variant *mainsec_DC_R1_plusUSreinst*, non-citable pending the R3 cross-model audit); the Sweden row is drawn from the OSOV reinstatement series (*b1-sweden-v2*). The table reports the second population of the correction, in which firms excluded from the baseline panel are reinstated and average $\bar{\kappa}$ is recomputed over the enlarged panel. The reference column is the panel average before reinstatement: for the United States it is the headline-series panel (the MAIN ∪ SECONDARY correction of the in-panel firms); for Sweden it is the baseline of the OSOV reinstatement series. The enlarged column is the average after reinstatement, and the delta-total column is their difference. The delta total is the sum of an incumbent-movement component and a composition component; the composition component is the implied residual, contributed by the reinstated firms' own κ, and the incumbent-movement component is the change borne by firms already in the panel. The incumbent-movement total decomposes into a dilution component (the denominator effect of enlarging the firm count) and a linkage component (new common-ownership links from reinstated firms to incumbents). For the United States, 946 excluded multiclass firms (representative CIK nodes) are reinstated (450 with full recomputed control vectors, 496 as dilution-only), moving the panel average from 0.3606 to 0.2301; the reinstated firms are dual-class-routed by construction, and the reinstatement engine applies no ownership-chain correction to any reinstated firm (the applied legs are recorded in the build manifest), so their ownership-chain layer has a construction-imposed value of zero and does not reflect measurement. Provenance and gate statistics are recorded in *reinstatement_numbers.json* and the *build_manifest_r15.json* manifest.

B Measurement of the unverified vendor-sourced tier (transitional)

This transitional appendix documents the measurement apparatus behind the unverified vendor-sourced tier that the headline series admits under benefit of the doubt; once the post-ship jurisdiction verification round (pending-work item 1) promotes or refutes the parked identities, the apparatus collapses into the record-grade ledger and this appendix is retired.

B.1 Construction detail

The passages below give the full construction mechanics behind the correction's three derivation legs and the China vehicle-level overlap, moved here from the main-text construction section (§3.2, §3.3, §3.4, and §3.6), whose paragraphs point here.

Control-chain reconstruction.

Deterministic parsing converts the vendor's percentage codes and holder-type codes to numeric stakes and typed owners. Line items are then consolidated into ultimate-owner blocks: individuals are aggregated into families under keys assigned offline by the judgment tier, corporate holders are rolled up to their global ultimate owner, sibling special-purpose vehicles are merged, and free float, treasury positions, and unnamed residuals are dropped from the block structure. A concert tier resolves, with cached verdicts, which legally distinct co-investors constitute a single control group and which are rivals to be left apart. Three identity passes supplement the base build. A named-beneficial look-through re-keys nominee register entries that carry their beneficial owner inline to that owner and drops the rest as float. Using the impossibility argument that one firm's named blocks summing to more than 100% of voting rights implies that at least two blocks overlap, a deduplication pass deterministically generates candidate same-entity pairs, each adjudicated in the judgment tier against web evidence (a translation variant, an acronym and its expansion, or a person and their personal holding company being folded into one owner, with the absorbed block's cash-flow leg re-attributed to the survivor), because the distinction between a genuine alias and a coincidental collision between unrelated institutions requires judgment. Throughout, structural guards protect the build from model failure: a rate alarm blocks any build in which the tier marks an implausible share of named individuals as dispersed, and population-level movement guards verify after each pass that country control shares moved only within tight bounds. Validation is a method-level comparison, because the vendor's current vintage cannot replicate Aminadav and Papaioannou's 2012 cross-section; the vintage gap is stated explicitly. Time-stable named pyramids provide a sharper check: the cascades behind Hermès, LVMH, and L'Oréal resolve to the correct controlling families with economically sensible wedges, near zero where the family wholly owns its holding chain and large where control rests on a thin cash-flow stake.

Non-US dual-class construction.

Construction proceeded in four stages, with the first mapping sources of per-holder and per-class disclosure in each jurisdiction, including registers maintained by securities regulators, notifications of threshold crossings, stock-exchange filings, and annual reports. Negative findings document a jurisdiction's absence from the panel, and every selected document was archived under a checksum manifest before parsing, making the build reproducible against the documents fetched. Deterministic, register-specific adapters then handle table walking, footnote linkage, locale and number-format handling, and PDF parsing, mapping each register into a common schema and routed by register archetype (structured machine-readable filings; semi-structured filings printing both ownership legs; non-Latin-script PDF regimes; EDGAR 20-F filings of foreign private issuers; notification streams carrying only the voting leg, joined to a separately sourced capital leg). Most non-US registers publish each holder's percentage of capital and percentage of votes side by side; the published figure is taken as primary, the recomputation from class terms provides a cross-check, and the divergence between the two becomes a quality instrument attached to every row. Two final checks apply to the committed panel. A hard universal bound excludes an entire firm whenever any computed leg leaves $[0, 1]$ or a wedge reaches one in absolute value, treating the violation as a parse or denominator error rather than a data point, and a statute-derived envelope, computed per register from the lawful vote-ratio cap and audited statute by statute, moves individual rows above the legal ceiling into review. Validation uses named controller anchors, including the Wallenberg sphere at SKF and Porsche SE at Volkswagen; verification against the primary register is repeated on every rebuild, with any movement treated as a regression. A round-trip gate required the generalised schema to reproduce the US-derived wedge set exactly before any foreign number was trusted. The remainder of the adjudicated candidates that carry no genuine wedge are equal-vote multiclass structures (dual listings, nationality classes, par tranches), instrument artifacts (warrants, preferred share lines, de-SPAC leftovers), board-seat rather than voting wedges, or firms with no extractable holder data.

United States DEF 14A derivation.

The universe is the excluded-issuer set bridged to EDGAR filings by CIK and collapsed to 946 representative CIK nodes: of these, 702 are in-scope multiple-class candidates, and the derivation resolves 360 genuine wedges, 261 equal-vote multiclass structures, and 66 single-class artifacts, leaving 15 documented as unresolvable. The extraction has three tiers. A deterministic layer resolves merged and spanning Item 403 headers, preserves each footnote's attachment to its cell, and anchors votes-per-share extraction to the narrative spans where it lives; a bounded language-model fallback decides the cases the deterministic layer cannot, returning output against a fixed schema with every numeric validated by pattern and range and frozen in a provenance-tagged cache, so the dataset remains deterministic; every value carries an extraction-method tag, and rows that fail even the fallback's validation escalate to a review flag. Two conventions are fixed in code. Voting power is emitted both as the issuer's as-reported percentages, which use holder-specific denominators and are not summable across holders, and as a single-denominator consistent measure suitable for aggregation, with the divergence between the two surfaced as a column. The wedge is computed at the consolidated control-group level, both legs built from the group's summed holdings before differencing, matching the convention of the dual-class literature; accuracy was assessed against a reconciled benchmark, with a set of golden-benchmark firms reserved for validation and excluded from the derivation entirely.

China vehicle-level overlap.

China is the closest overlap case, because the co-authors consolidate state holders under one government owner while the reconstruction traces the same holders through provincial vehicles. There the discovered mass is 58.8% quarantined, 41.0% replacement, and 0.16% added, and the cross-model overlap audit judged the residual state-vehicle matches indeterminate, holding the eight arithmetically overfull rows for manual adjudication instead of demoting them automatically.

B.2 Record-grade definitions

We grade every candidate record by the strength of the evidence behind it into three tiers. The verified discovered controllers (internally MAIN) are 165 anchors, each mapped to a named row in the hand-coded Thomson Reuters roster, individually reviewed, and certified through repeated adversarial cross-model audits; a documented ten-record spot check of these anchors against primary filings returned no errors (one-sided 95% upper bound 25.9%). The unverified vendor-sourced controllers (internally SECONDARY) are 1,352 records taken from the ownership vendor's chain data and confirmed only where web evidence permitted; a fresh adversarial sample of these records, checked against local-language primary filings, fails at a measured rate of 26.0% (one-sided 95% upper bound 38.1%), so the tier is admitted under benefit of the doubt with its error disclosed. The remaining 45,838 candidate records in the 2024Q4 tier output (the full applied ledger at that quarter is 165 / 1,352 / 45,838) are set aside as identities the roster could not corroborate (internally QUARANTINE). The quarantined tier divides into refuted and unresolved portions: about 38% of its discovered-controller denominator mass is affirmatively contradicted by a primary source, while the remaining 62% is unresolved and parked for want of corroborating evidence, without having been shown wrong.

B.3 Measured quality of the unverified vendor-sourced records

Table 24 documents the disclosed error rate behind the unverified tier. It reports a verification census of the web-checked records, an error taxonomy of the demotions, and a fresh local-language recheck of surviving unverified records. The recheck provides the 26.0% figure quoted in the main-text attribution section.

Table 24: Vendor-sourced control records: measured quality

Panel A. Verification census

Records web-checked against primary sources	1,656
Affirmed against a primary source (kept)	786
Thin / low-confidence evidence (kept)	516
Demoted — identity not supported	354

Panel B. Error taxonomy of demoted records

Error class	Detection mechanism	Count	Example
False-person attribution	Primary filing lists no such holder	178	Lim Ah Cheng / Stamford Land (SG)
Superseded controller	Filing shows the controller divested	47	Icahn / Cheniere: exited 2023
Custodian / manager as controller	A manager's rolled-up position, not personal	39	McBirney / Topline Capital (US)
State-institution rollup	State-institution block → sovereign	28	EPF / ViTrox (pension block)
Wrong-issuer link	Holder belongs to a different issuer	19	Malayandi / SMRT Holdings (MY)
Sovereign custodial rollup	Sovereign rolled up across micro-caps	14	Government of China / D&O (MY)
Sovereign chain artifact	Chain ends spuriously at a sovereign	9	Government of France / Catana (FR)
Corrupt identity key	Malformed identity as a controller	6	Johnson Matthey keyed as MATTHEY
Activist-fund principal	Fund / LP capital, not personal	5	Shulkin / Brainsway (Valor Equity)
Deceased ultimate owner	Owner deceased; not succeeded	5	Enderle / City Lodge (deceased 2019)
Duplicate-alias key	Same holder recorded under two keys	3	Prommasutra vs Phrommasut, one holder (TH)
Public authority as controller	A government body, not an owner	1	Legislative Assembly / SPAC (US)
Total demoted		354	

Panel C. Measured residual error rate

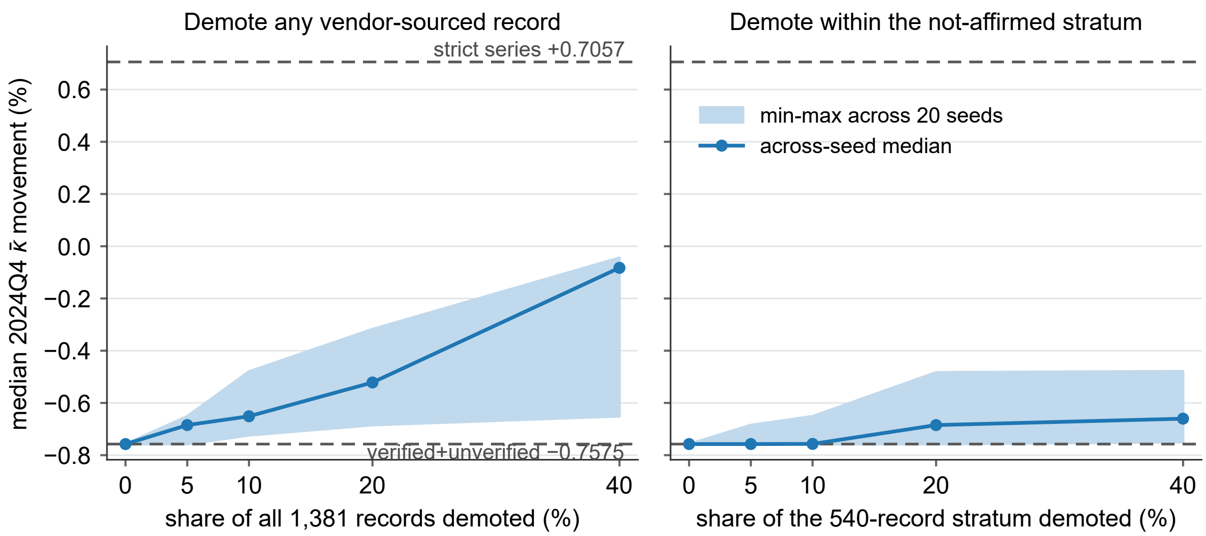
Unverified vendor-sourced controllers sampled	50	records
Failed vs primary filing	13	(26.00%)
One-sided 95% upper bound		38.13%
Verified discovered controllers sampled	10	records
Failed vs primary filing	0	(0.00%)
One-sided 95% upper bound		25.89%

Note: *Panel A* reports the round-9 verification census: of 1,656 owner–company records web-checked against primary sources, 354 were demoted to the quarantined tier as unsupported identities (including three duplicate-alias keys a same-issuer, same-stake scan resolved to a single holder) and 1,302 were kept. Of those, 786 were affirmed against a primary source and 516 had thin or low-confidence evidence, with the evidence-faithful split reclassifying as low-confidence 212 keeps the assembler had recorded as web-affirmed. *Panel B* classifies the demoted records by error class, with census counts and one demoted record as an example of each. *Panel C* rechecks a fresh sample of 50 surviving unverified records against local-language primary filings (for example SSE/cninfo, EDINET, or the annual report): 13 failed, a 26.00% error rate with exact one-sided 95% upper bound 38.13%, while 10 verified discovered controllers returned no failures. The census counts predate the R15 netting layer, which subsequently demoted a further 8 of these records to the quarantined tier (the overlap-audit quarantines) and, through the arithmetic backstop, removed 21 more from the 2024Q4 panel, bringing the active unverified tier from 1,381 records to 1,352. The panels measure the vendor data's quality and show that unverified vendor ownership records cannot be taken at face value.

B.4 Stress robustness of the headline under vendor error

A simulation tests whether the headline's control-chain leg remains within its graded bracket when unverified records fail at the measured error rate of 26% and are removed. Figure 8 demotes random shares of the unverified vendor-sourced records up to 40%, beyond the measured upper bound, and recomputes the median movement at each rate.

Figure 8: Stress robustness of the headline's control-chain leg under random demotion



Note: Each panel plots the all-49 median 2024Q4 $\bar{\kappa}$ movement of the verified + unverified vendor-sourced series as a random share of the unverified vendor-sourced records is demoted to the quarantined tier. The two panels demote from different pools: the left panel removes a random share of all 1,381 web-checked unverified records, whereas the right panel restricts demotion to the 540-record stratum not affirmed against a primary source, so its 5, 10, 20, and 40% correspond to 27, 54, 108, and 216 records, equal to 1.96, 3.91, 7.82, and 15.64% of the full 1,381. The shaded band spans the minimum to maximum across 20 fixed seeds at each rate and the marker is the across-seed median; the dashed lines are the graded bracket, the strict series (+0.7057%) above and the verified + unverified base (−0.7575%) below. Every draw remains inside the bracket and moves toward the strict series as more records are removed, so no demotion rate the measured error could imply pushes the series outside the graded bracket. The 8 records the R15 overlap audit demoted are a realized subset of this stress.